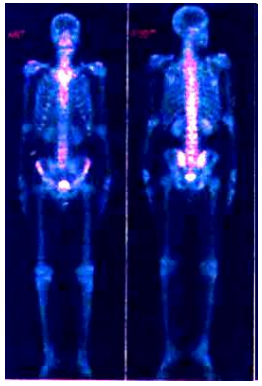
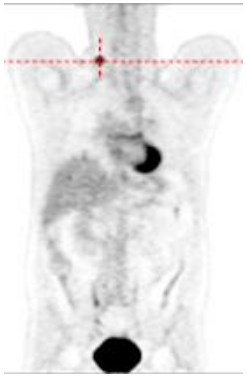
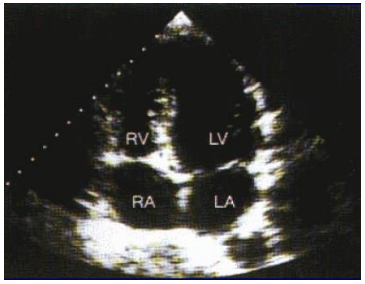
Medical imaging modalities I

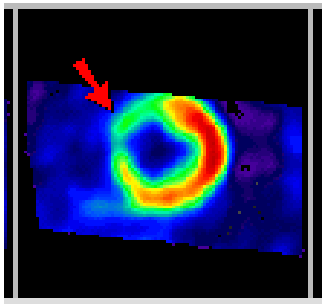
Mónica Abella (monica.abella@uc3m.es)

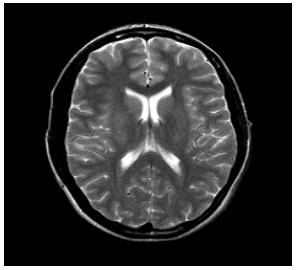




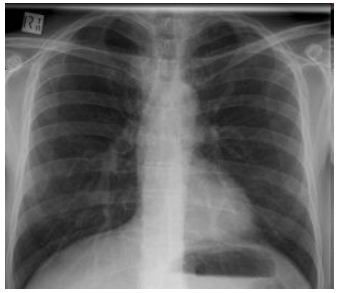


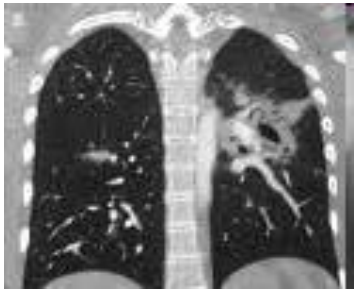


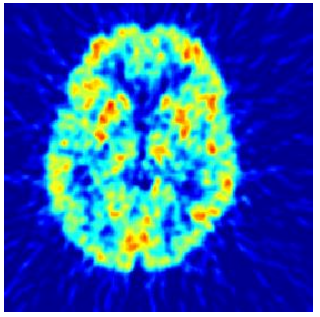






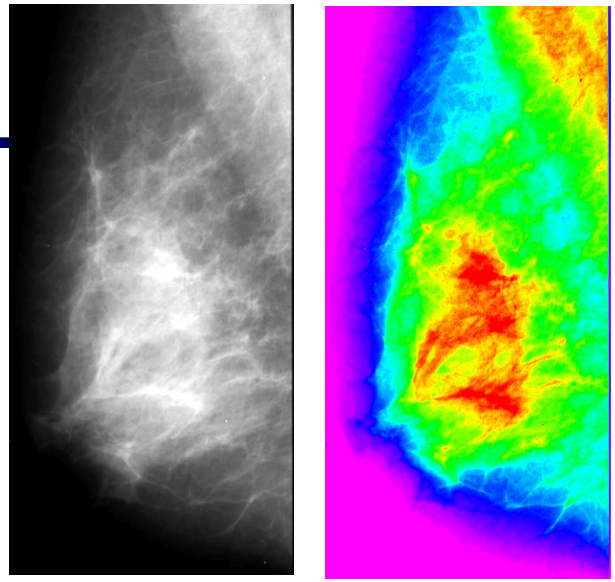




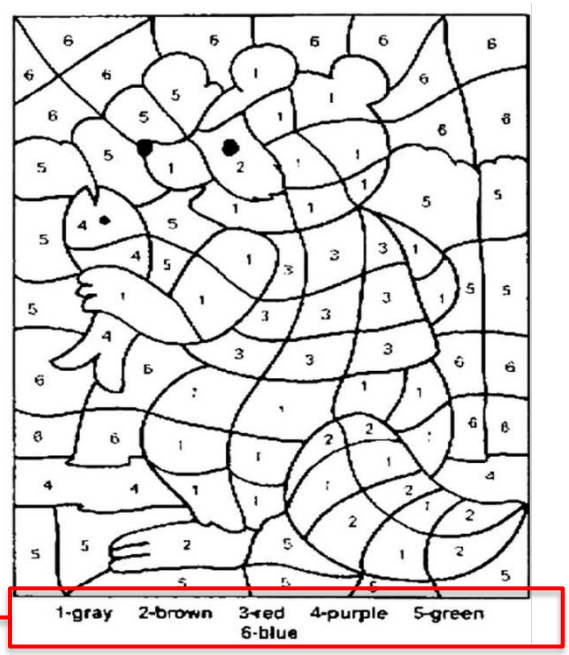


Medical image

- "Map" of some physical or chemical property in the organism, numbers not a picture!



0	1	1	1	0	0	0	0	1	0
0	1	1	1	0	0	0	1	1	0
0	1	1	1	0	0	0	0	1	0
0	1	2	4	0	7	7	2	1	0
0	1	2	5	0	8	8	3	1	0
0	1	2	4	0	8	8	2	1	0
0	1	2	2	0	2	2	2	1	0
0	1	2	5	0	5	5	1	1	0
0	1	2	5	0	2	2	2	1	0
0	1	1	1	0	1	1	1	1	0

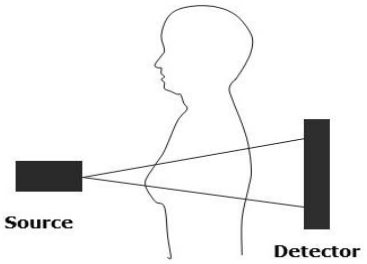


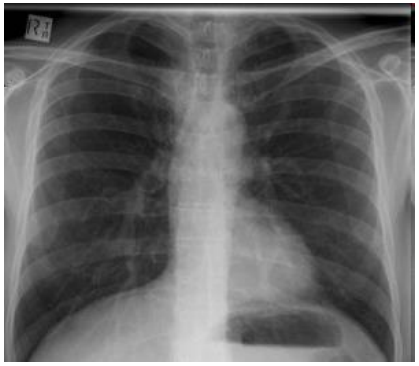
Look-up table (LUT)

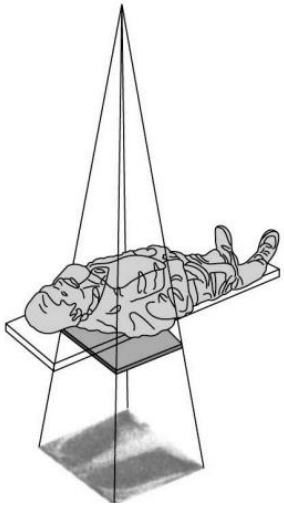
Projection vs. tomography

Projection

A "shadow" of the body in a particular direction





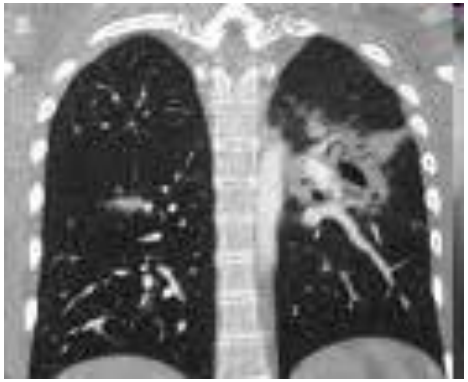


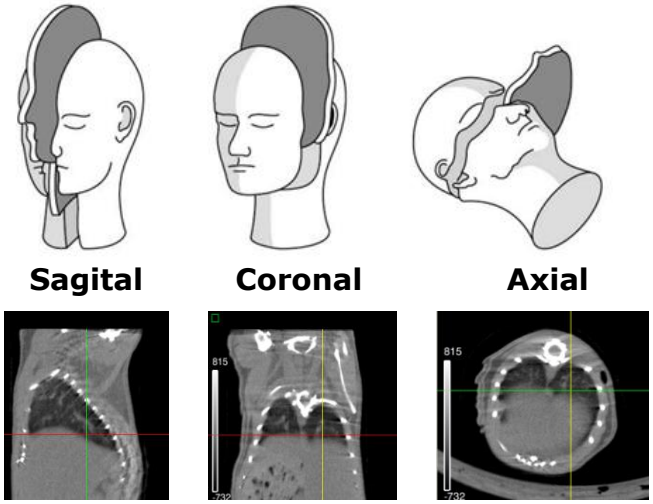
Tomography

From Greek, "tomé": to cut, "grafos": picture

We have depth information



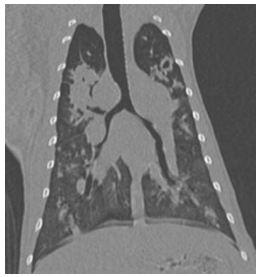




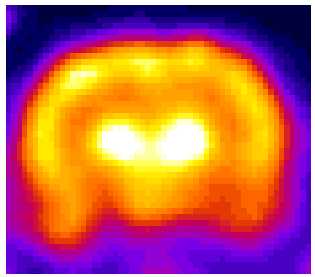
Medical image

- "Map" are a map of some physical or chemical property in the organism, numbers not a picture!

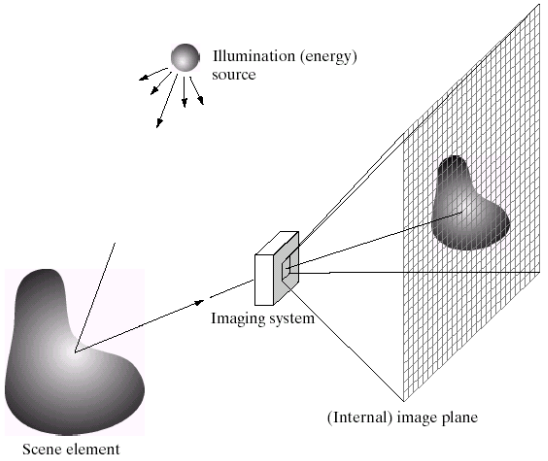
X-ray



Nuclear medicine

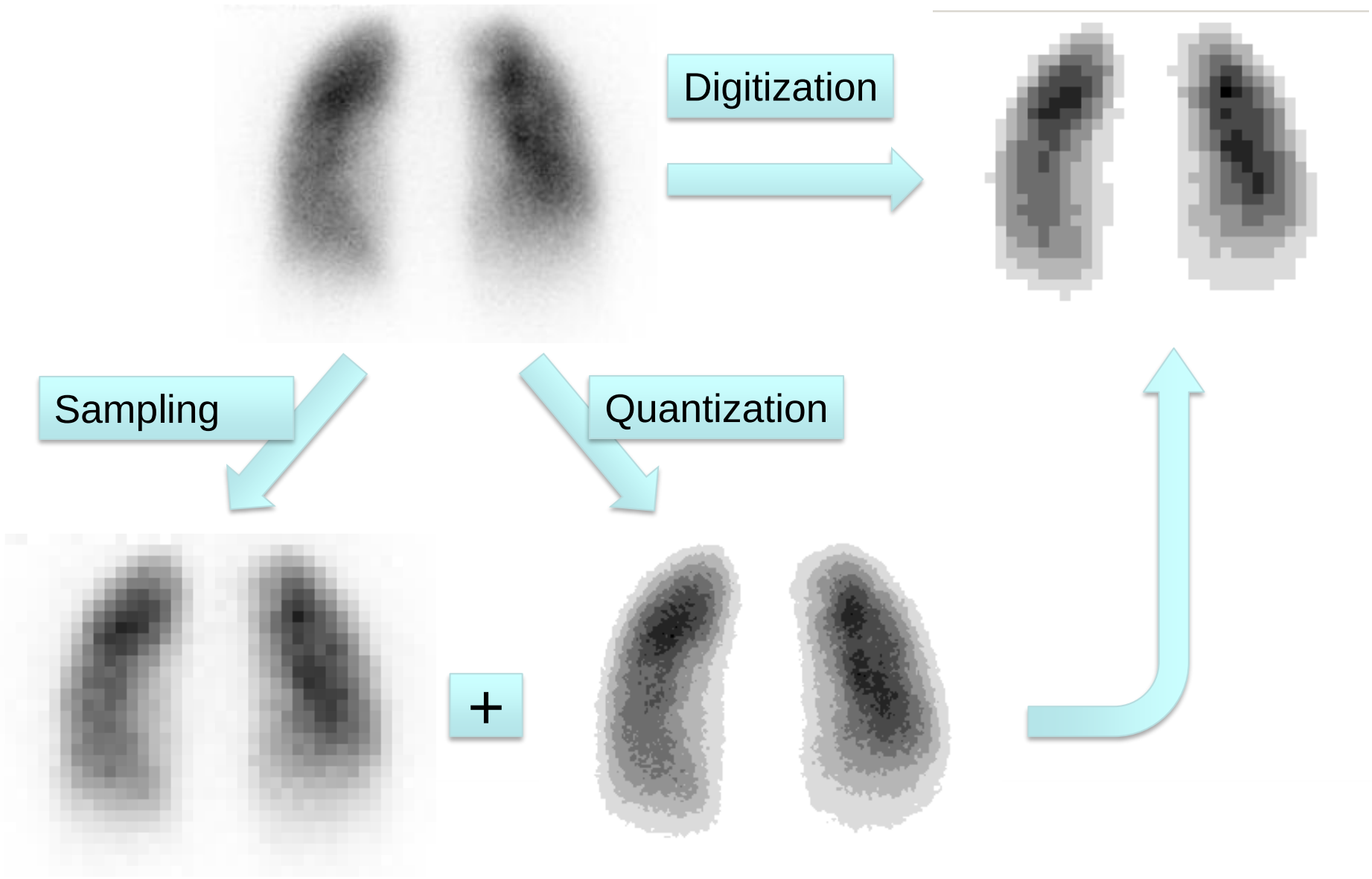


- Medical image acquisition involves the irradiation of the sample with a certain energy
  - Energy able to penetrate the human body and interact with internal body structures
  - Interaction depending on the physical or chemical property of tissues
  - Energy coming from the sample will land on a sensor material responsive to that type of energy and this generates an electrical signal



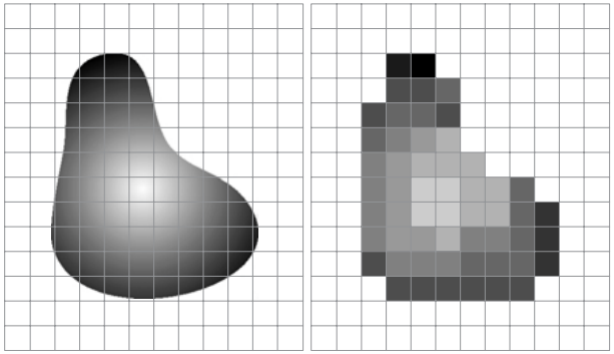
Different acquisition techniques: Modalities

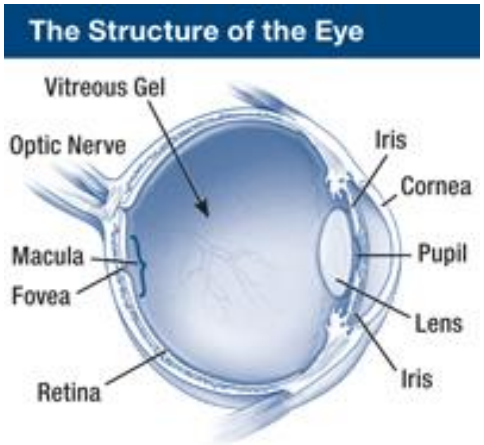
Digitization

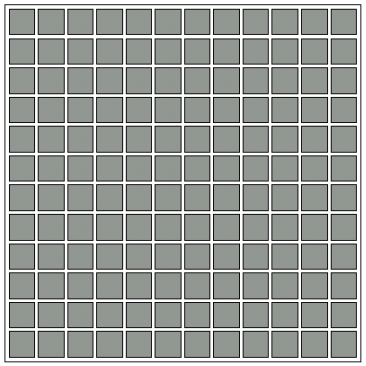


Digital imaging systems

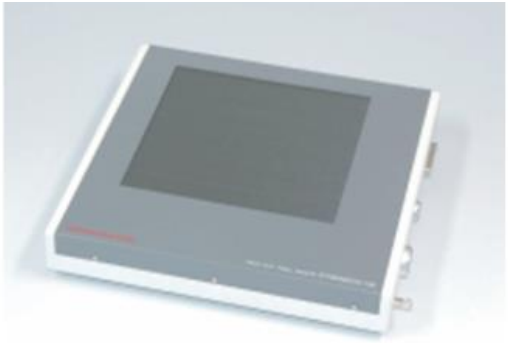
- Where do you think the sampling is happening?







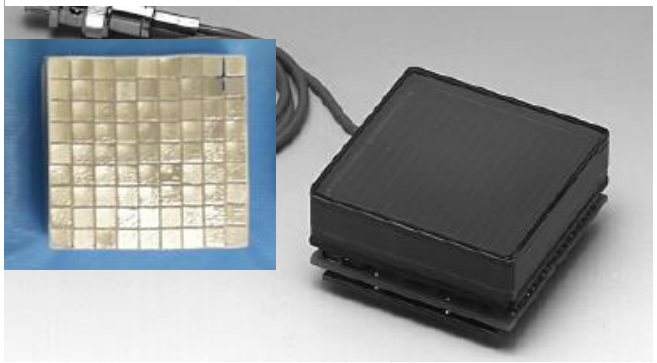
Flat panel (2000x2000 elements)



Digital camera detector (4000x3000 elements)



PET detector (8x8 elements)

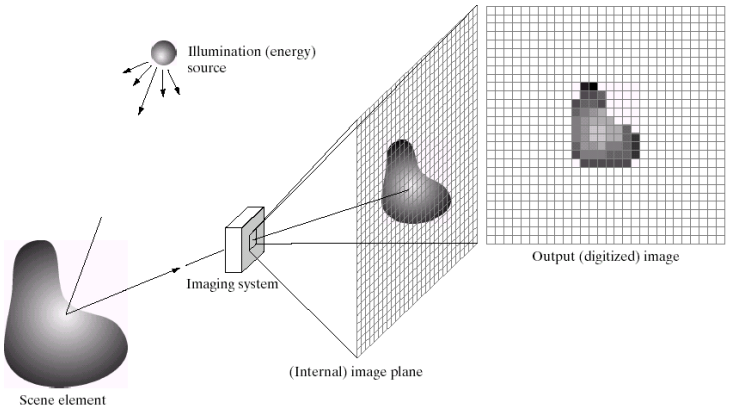


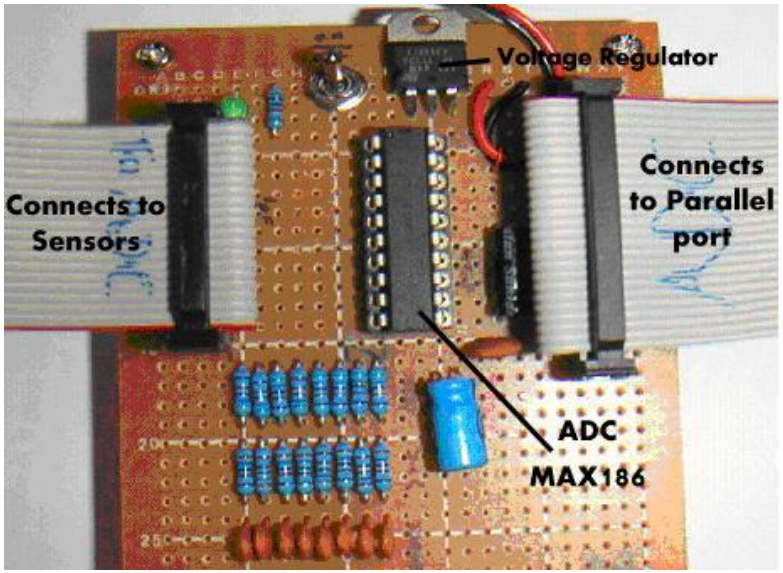


2D US transducers

Digital imaging systems

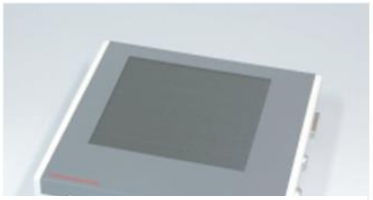
- Where do you think the quantization is happening?



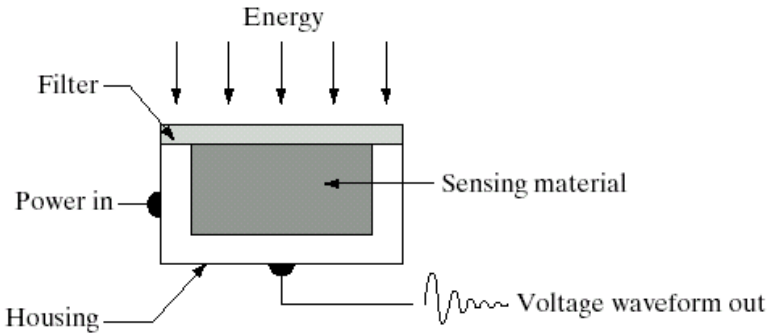


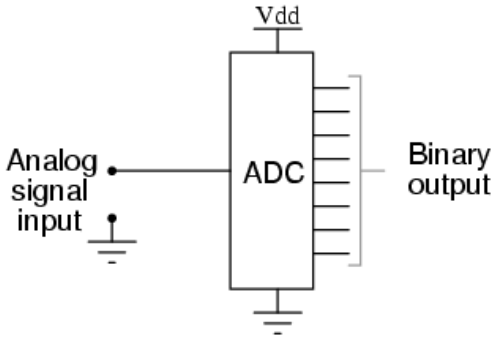


PET detector (8x8 elements)



Flat panel (2000x2000 elements)

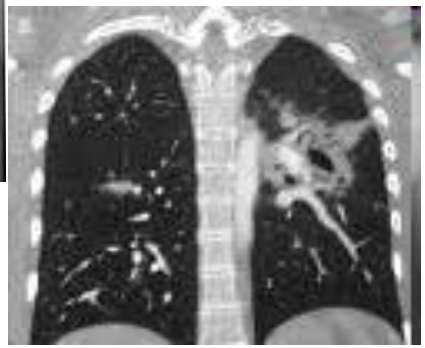




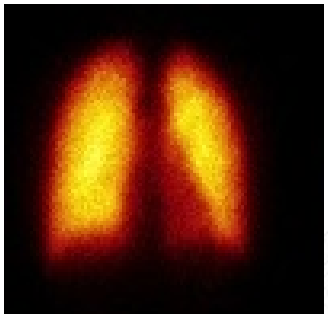
Main imaging modalities

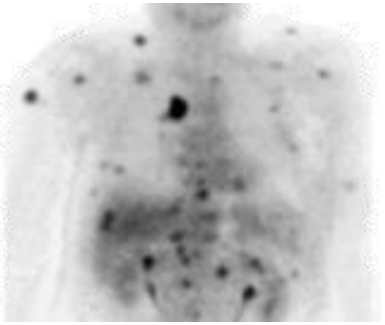
- X-ray imaging



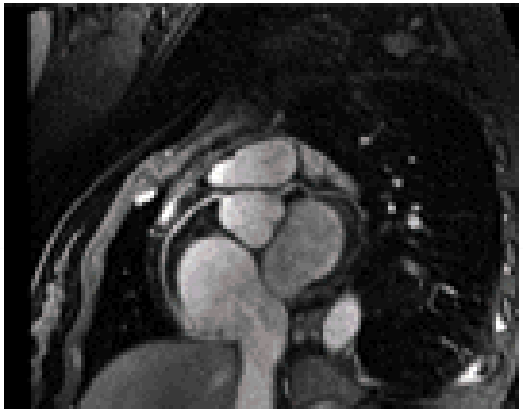


- Nuclear Medicine

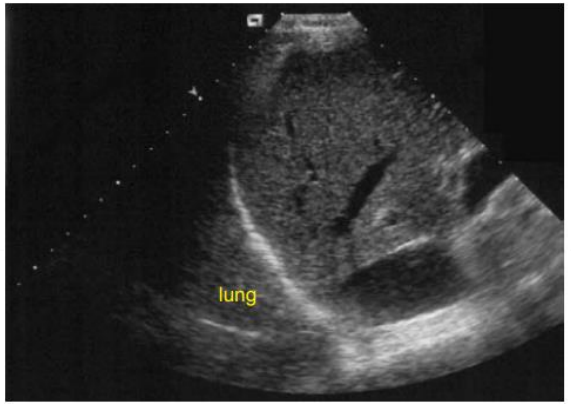




- MRI



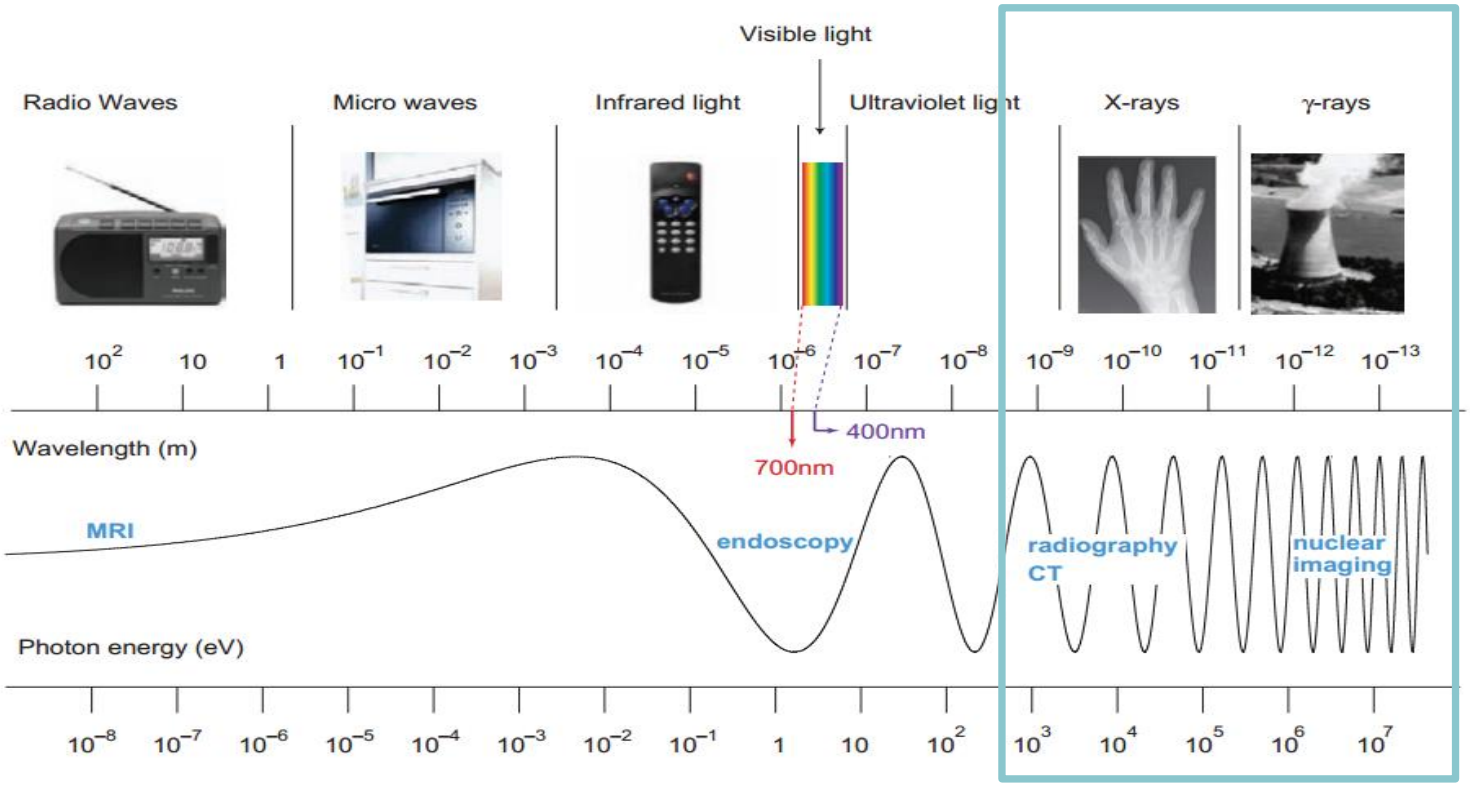
- Ultrasound



X-ray and nuclear medicine

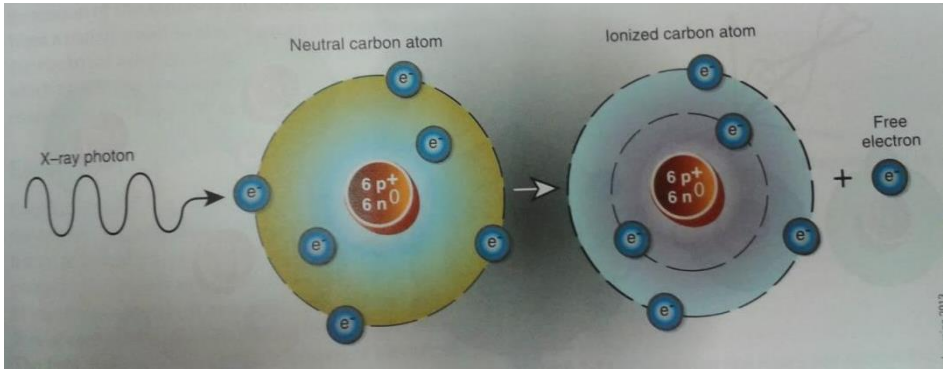
Energy in X-ray and Nuclear medicine

- Type of energy: electromagnetic radiation
  - X-ray imaging: X-ray photons
  - Nuclear Medicine: Gamma photons
- High energy photons: ionizing radiation

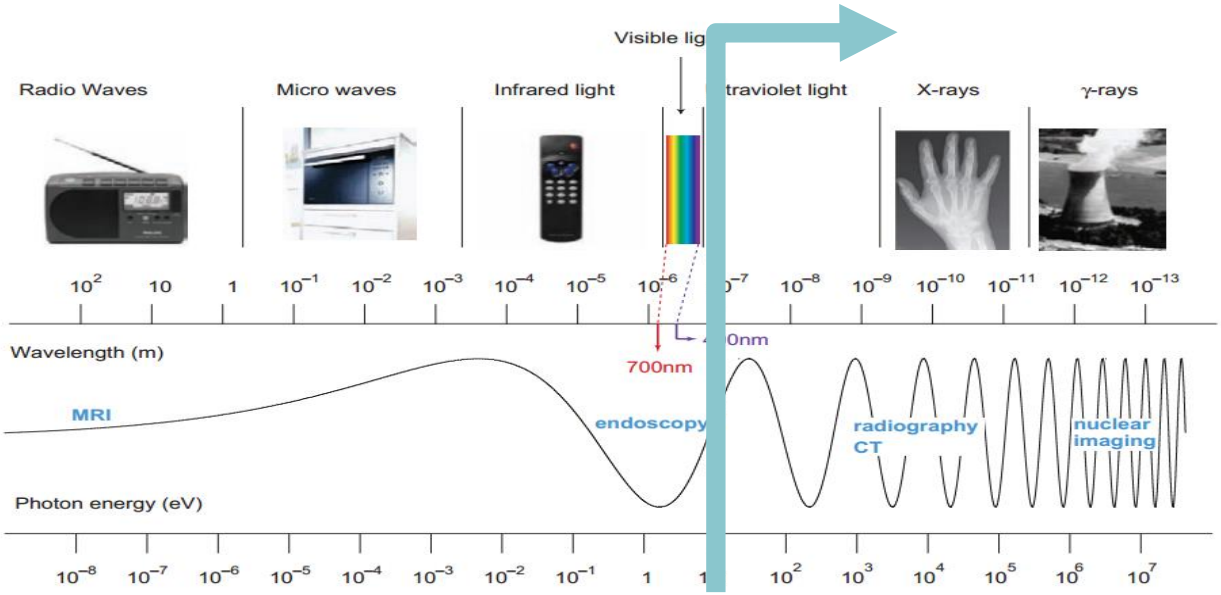


Energy in X-ray and Nuclear medicine

- High energy photons (> energy in the inner layers of atoms)
  - Able to strip electrons
  - Ionization: Formation of an ionic pair

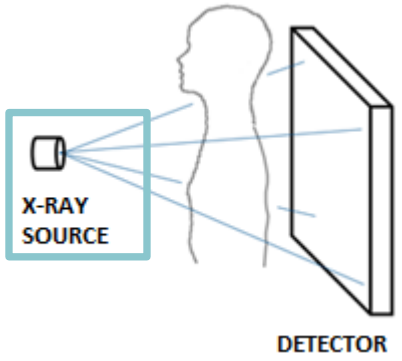


- Many elements in tissue < 1 keV (O: 0.5 keV, H: 13.6 eV)

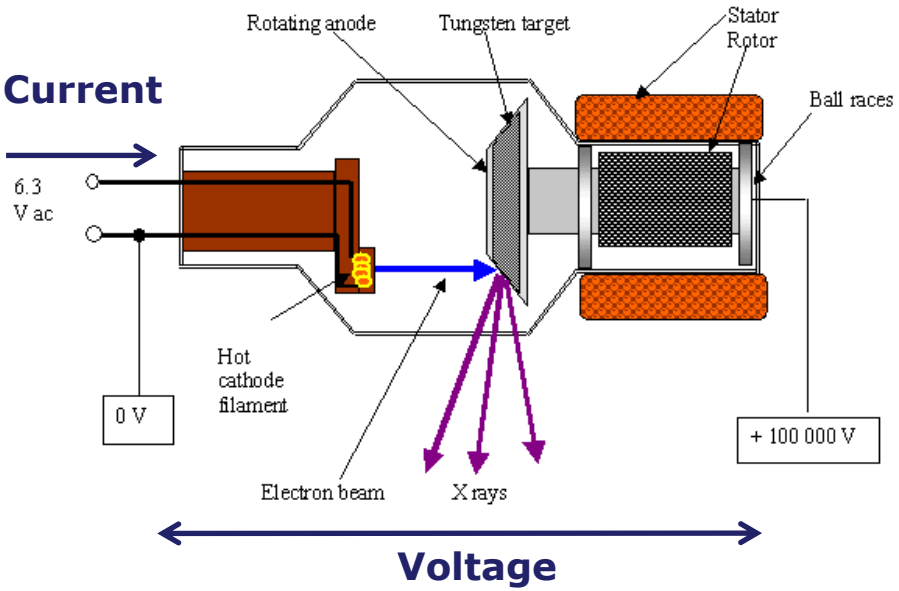


Source of energy

- X-ray: X-ray tube

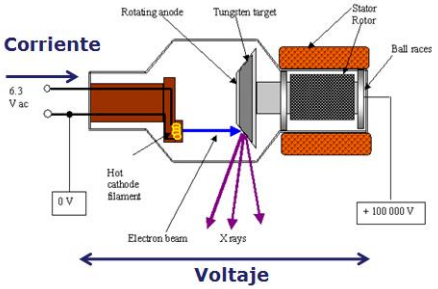




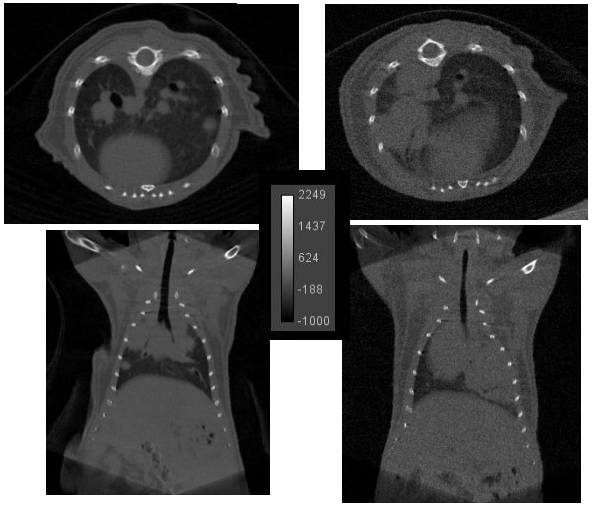


- Vacuum tube
  - The filament in the cathode is heated by a current and electrons are emitted (thermionic emission)
  - High voltage between cathode and anode accelerates the electrons towards the anode where their energy is converted into x-ray photons
- Most part is converted into heat, only 1% as X-ray photons

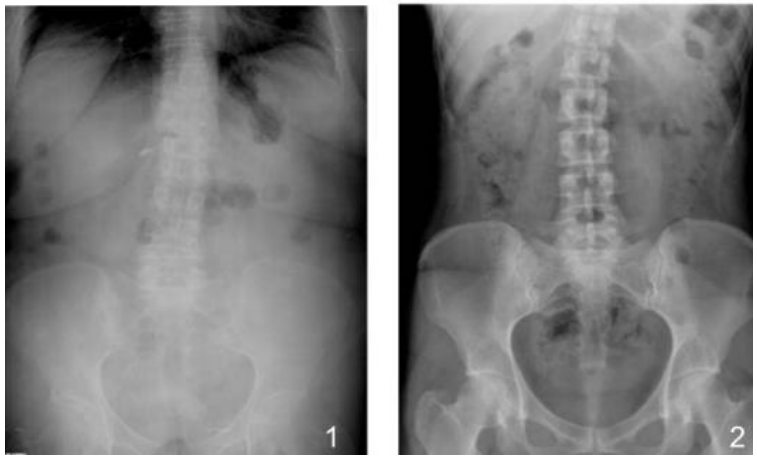
Source of energy in X-ray imaging



- Current
  - Number of photons
  - Better image quality (better SNR) but more ionizing dose
  - Typical values:
    - Clinics: mA
    - Small animal: µA

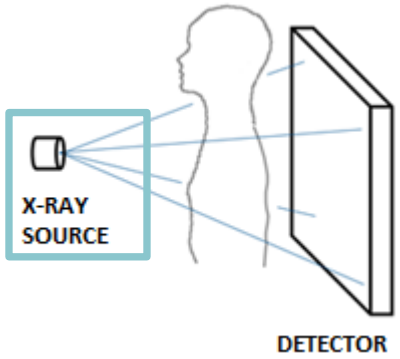


- Voltage: “hardness” of radiation
  - More penetration
  - Optimum: better contrast
  - Typical values
    - Clinics: 80-140 kVp
    - Small animal: 30-50 kVp

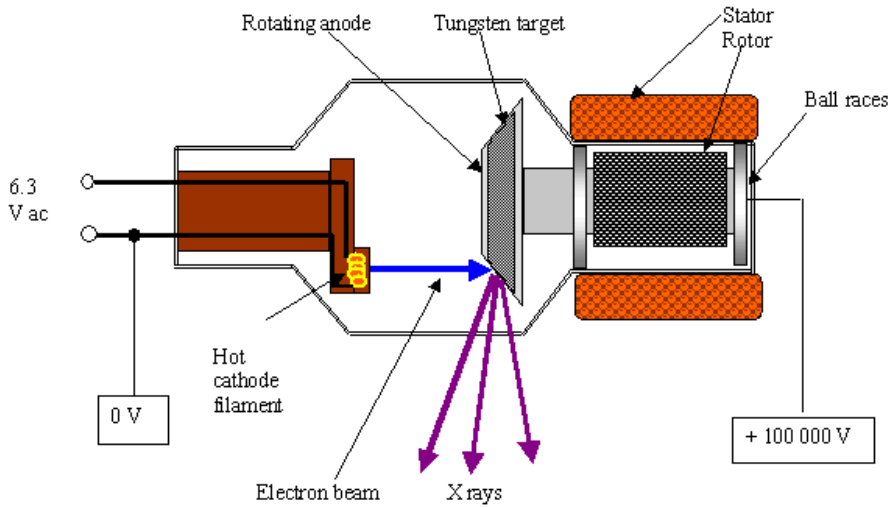


Source of energy

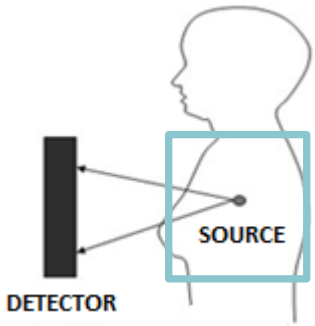
- X-ray: X-ray tube



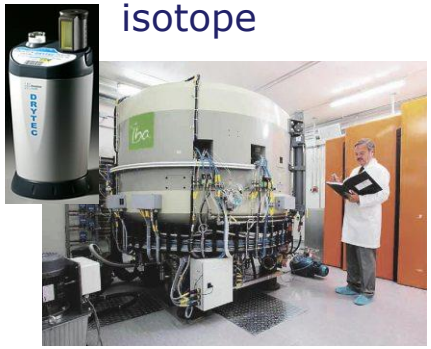


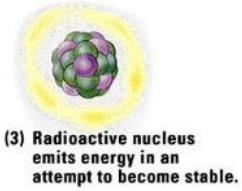


- Nuclear medicine: Radiotracer

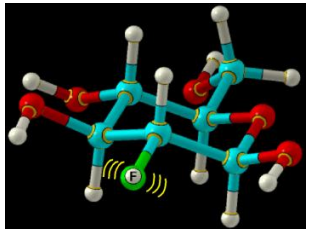


Radioactive isotope

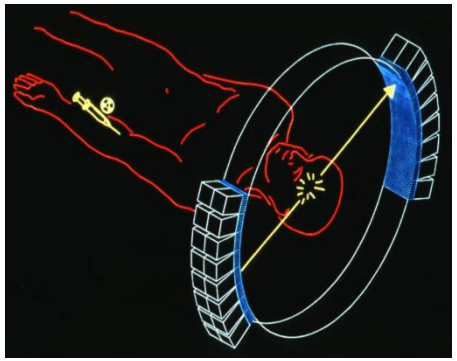




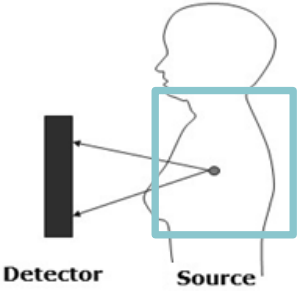
radiotracer



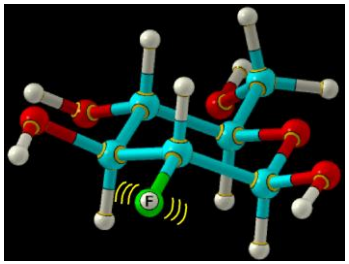


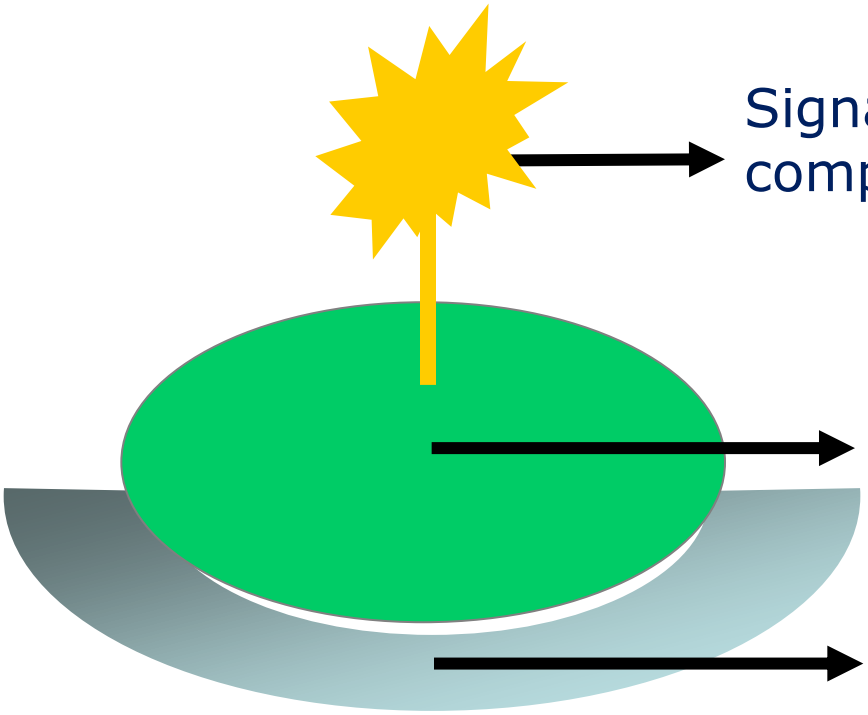


Source of energy in Nuclear Medicine

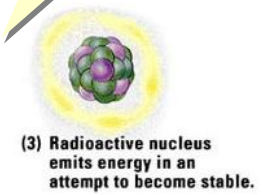


- Radiotracer:
  - Labeled compound
  - Incorporated to biochemical pathways
  - Detectable from outside
  - Trace concentration: does not alter the system





Radioisotope

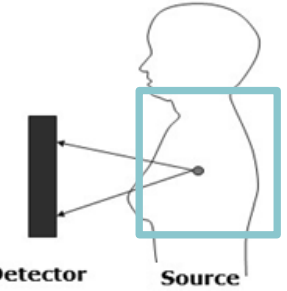




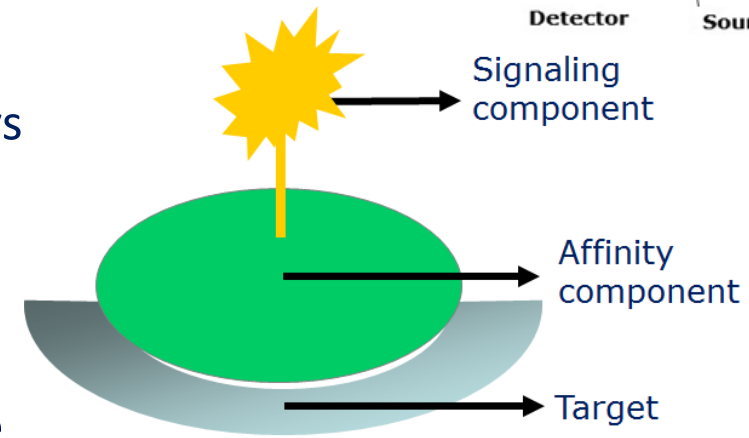
Determines the biological behavior (binding)

What biological process we want to study?

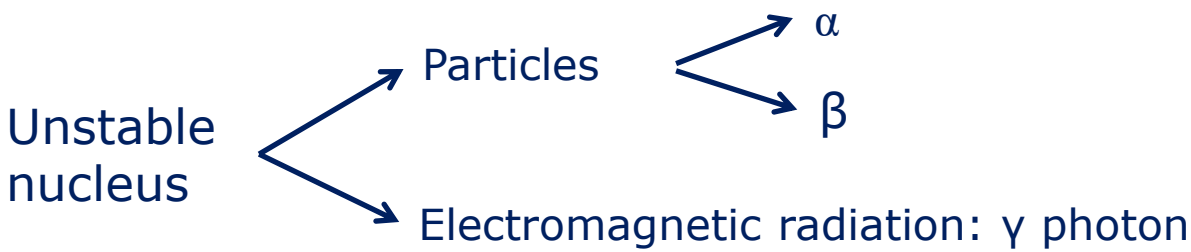
Source in Nuclear Medicine



- Radiotracer:
  - Labeled compound
  - Incorporated to biochemical pathways
  - Detectable from outside
  - Trace concentration: does not alter the system



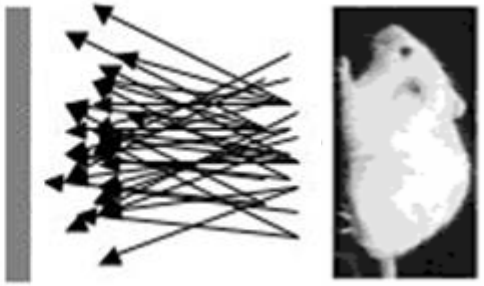
- Radioisotope: radioactive isotope
  - Radioactive decay: Spontaneous event for which a nucleus is transformed into another more stable emitting energy



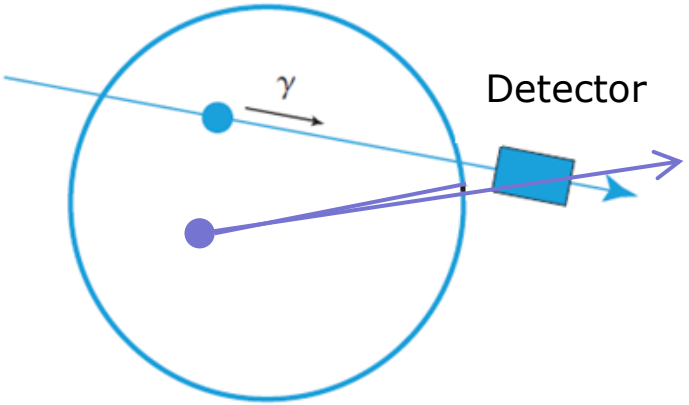
  - Activity: disintegrations per second (Bq/Ci)
    - Semi-disintegration period: time to decay to 1/2

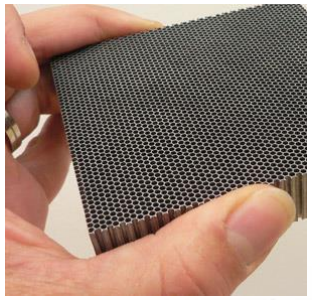
Collimation in MN

- Decay by gamma emission: one photon
  - **Scintigraphy and Single Photon Emission Tomography (SPECT)**

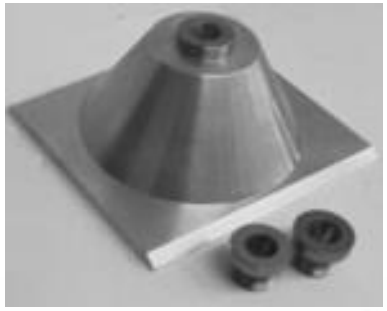


Hardware collimation

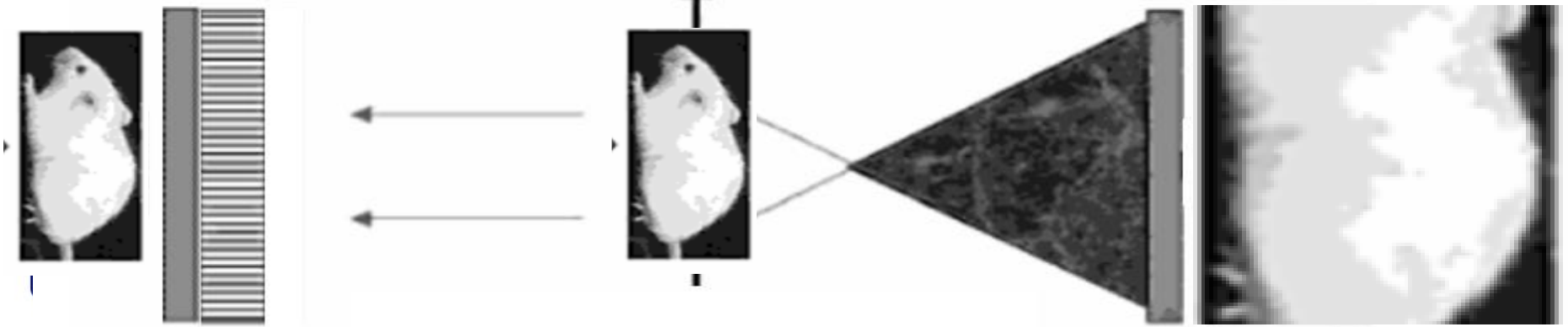




Parallel collimator

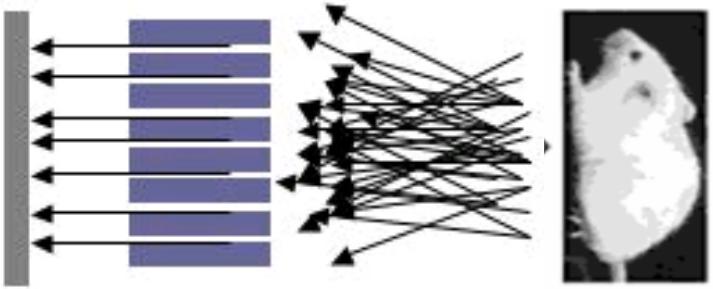


Pinhole collimator

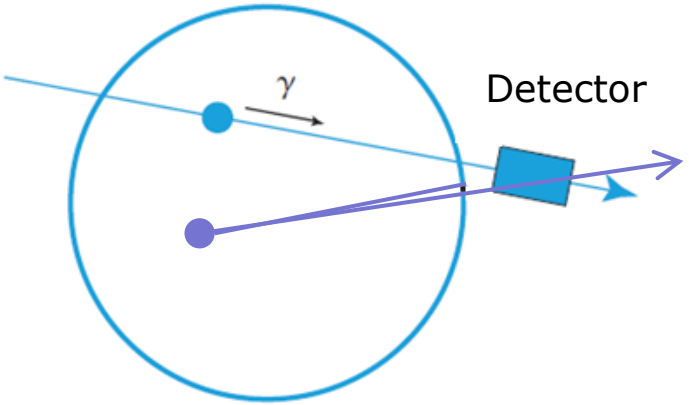


Collimation in MN

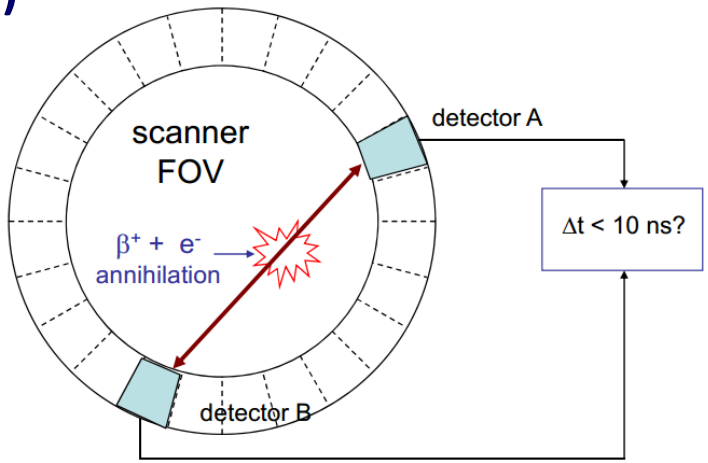
- Decay by gamma emission: one photon
  - **Scintigraphy and Single Photon Emission Tomography (SPECT)**



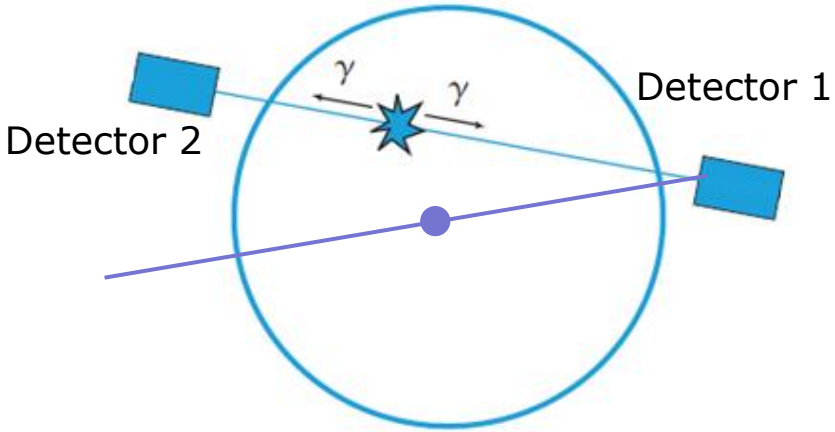
Hardware collimation



- Decay by positron emission (β+) : two gamma photons in opposite directions
  - **Positron Emission Tomography (PET)**

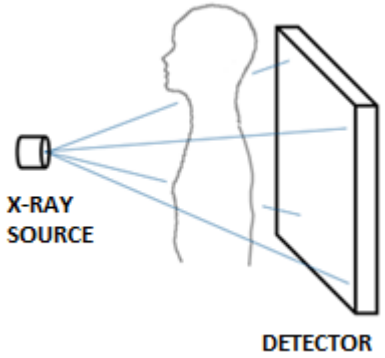


Electronic collimation

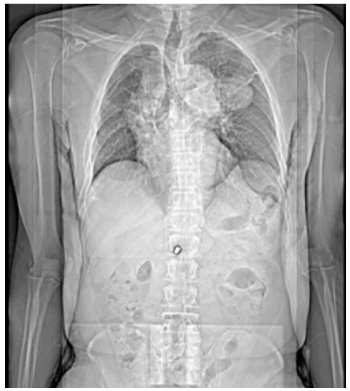


Projective systems

X-ray imaging

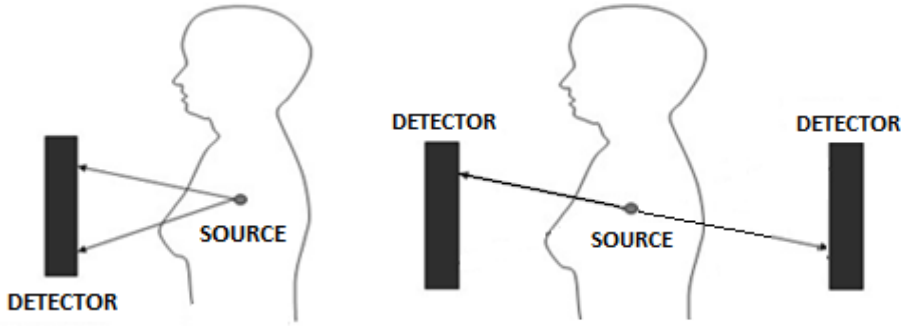


X-rays traverse the sample suffering different attenuations and are registered in the detector

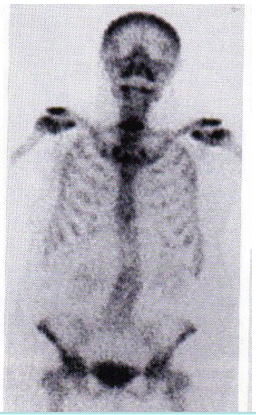


Map of attenuation properties of traversed tissues to X-rays

Nuclear medicine: SPECT/PET



Tracer emits gamma photons that are registered in the detector



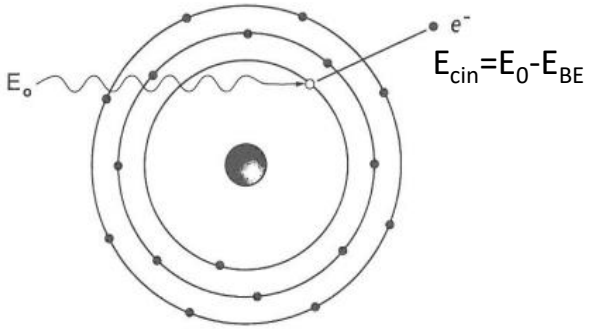
Map of tracer concentration

Projection image
The body is “collapsed” into an image (overlaid information)

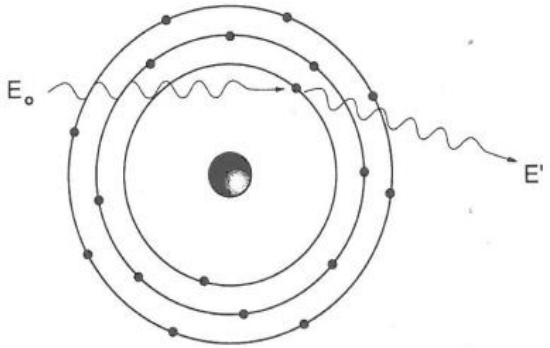
Interactions with the tissue

$$\text{Measured photons} = \frac{\text{Emitted photons}}{\text{Absorption in tissue}}$$

- Photoelectric effect:
  - The photon is completely absorbed releasing an electron (ionization)



- Compton scatter:
  - The photon looses part of its energy and changes trajectory



Origin of the contrast in the image

- X-ray imaging: Map of attenuation to X-rays of traversed tissues

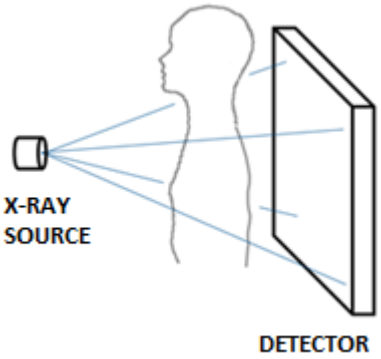
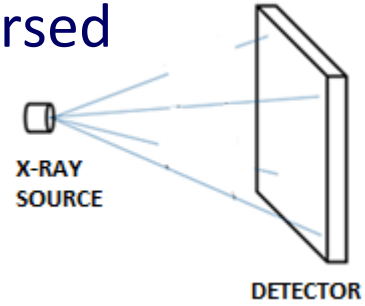
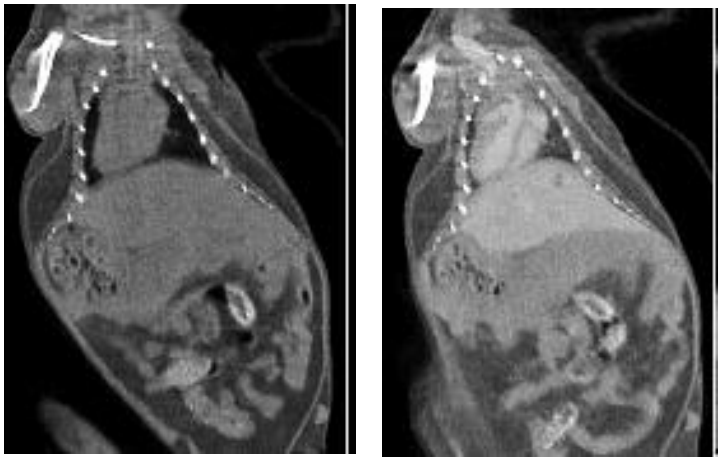


Image without sample



$$\text{Measured photons} = \frac{\text{Emitted photons}}{\textbf{Absorption in tissue}}$$

- Little difference of attenuation properties t X-rays between different soft tissues
  - Contrast agents: High attenuation to X-rays



Contrast agents in X-ray



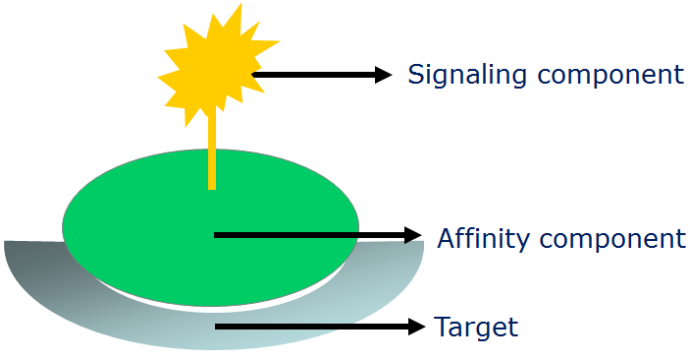


- X-ray imaging has poor contrast in soft tissue
  - Contrast agents: High attenuation to X-rays
- Common contrast agents:
  - Iodine (blood vessels)
  - Barium (digestive tube)
- Administration
  - Intravenous injection (blood vessels)
  - Oral administration (abdomen and pelvis)
  - Rectal administration (large intestines and other organs in the pelvis)





Is it a tracer?



Origin of the contrast in the image

- X-ray imaging: Map of attenuation to X-rays of trave tissues

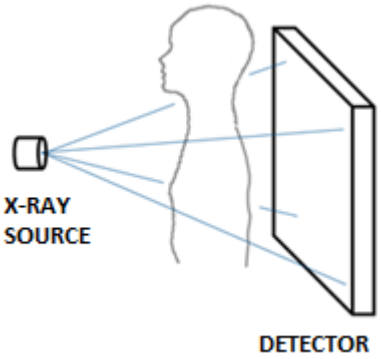
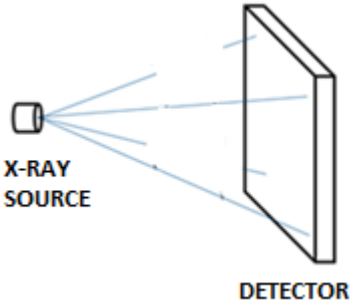


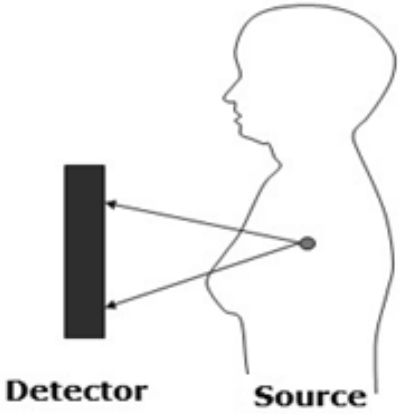
Image without sample



$$\text{Measured photons} = \frac{\text{Emitted photons}}{\textbf{Absorption in tissue}}$$

- Nuclear medicine: Map of tracer concentration

$$\text{Measured photons} = \frac{\textbf{Emitted photons}}{\text{Absorption in tissue}}$$



Detector

- Incoming energy lands on a sensor material responsive to that type of energy and this generates an electrical signal (generally, proportional to the integral of the incoming energy)

gamma/X-ray photons

↓ ↓ ↓ ↓ ↓ ↓

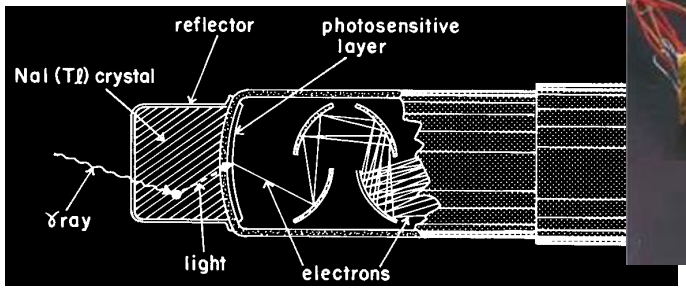
Scintillator: converts high energy photons into light photons

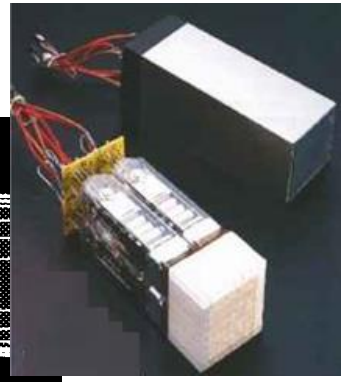
Photodetector: converts light photons into electric signal

↓

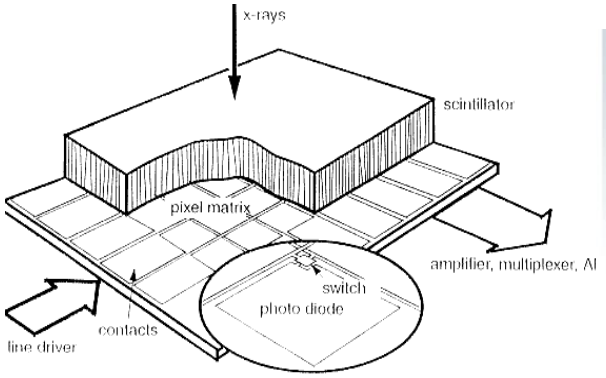
Measurement proportional to incoming energy

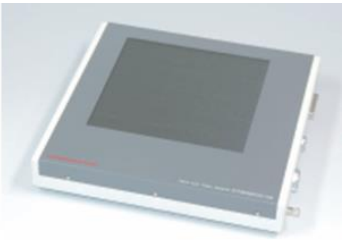
- MN: PET/SPECT





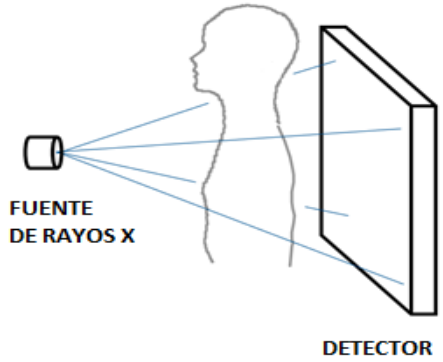
- X-ray imaging



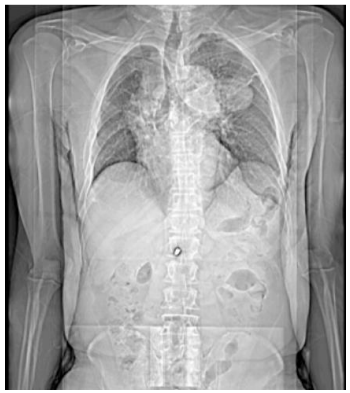


Projective systems

X-ray imaging

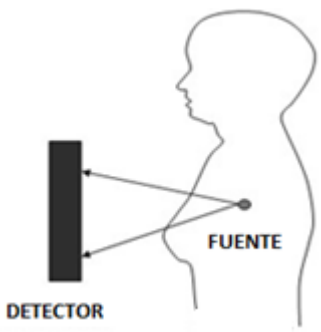


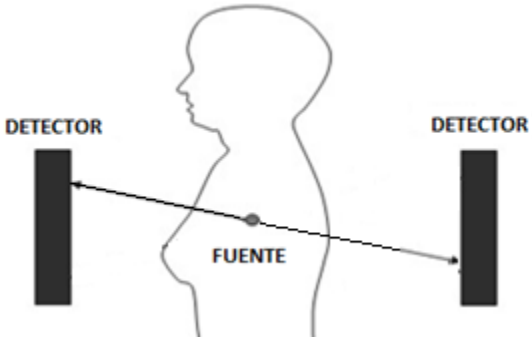
X-rays traverse the sample suffering different attenuations and are registered in the detector



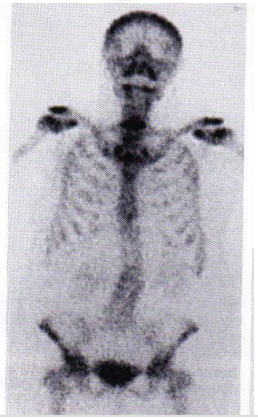
Map of attenuation properties of traversed tissues to X-rays

Nuclear medicine: SPECT/PET





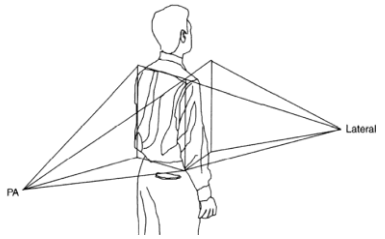
Tracer emits gamma photons that are registered in the detector



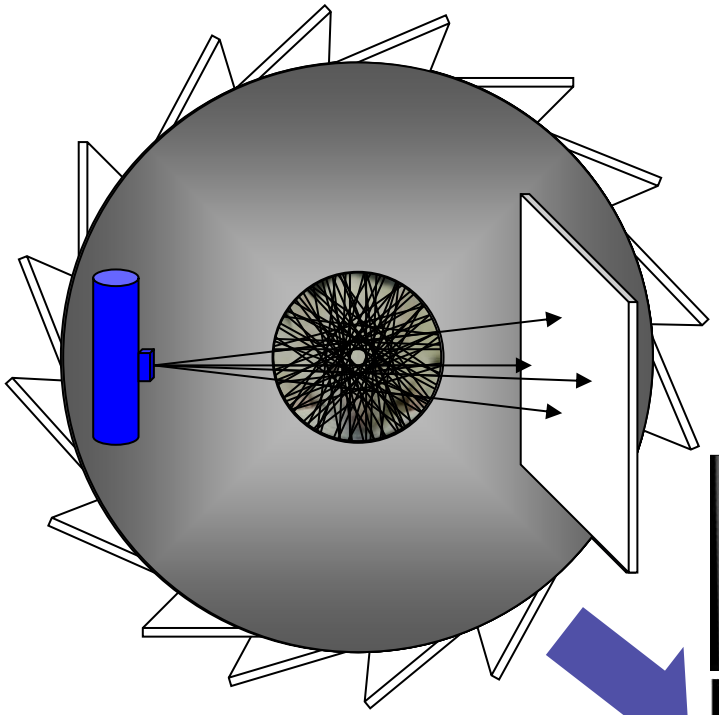
Map of tracer concentration

Projection image
The body is “collapsed” into a projection (overlaid information)

Computed tomography (CT)



- Godfrey Hounsfield, 1970: Godfrey Hounsfield, 1970: what if I take several radiographies from different views and combine the information?



Acquisition

Set of projections

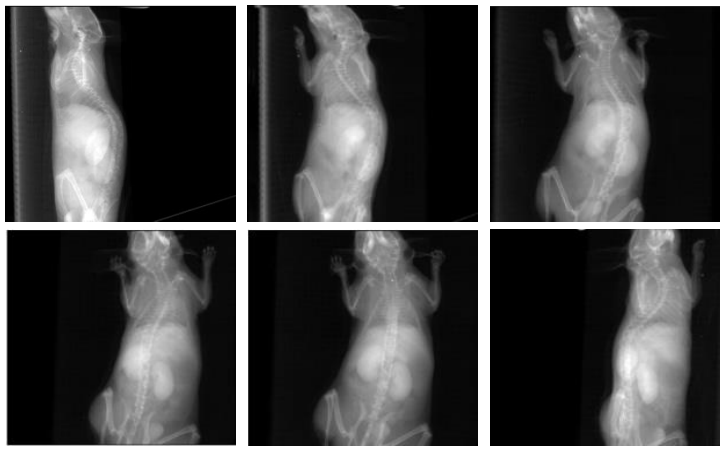
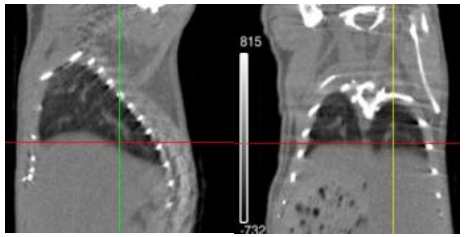
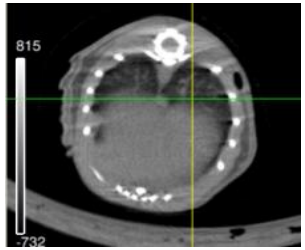


Image reconstruction

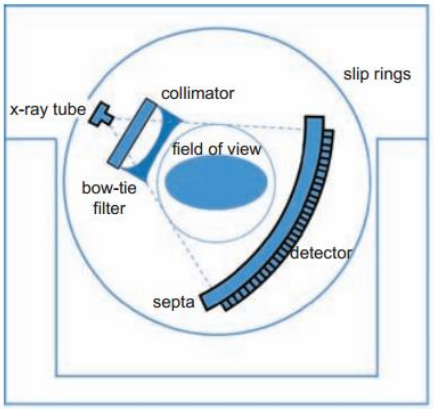
Map of attenuation to X-ray

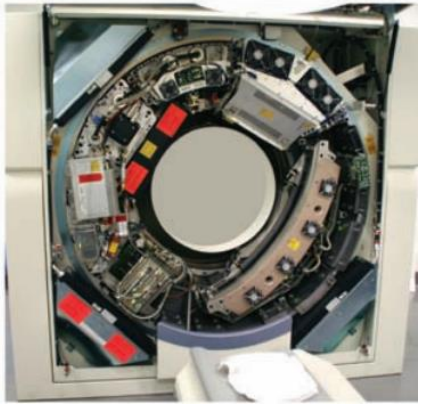


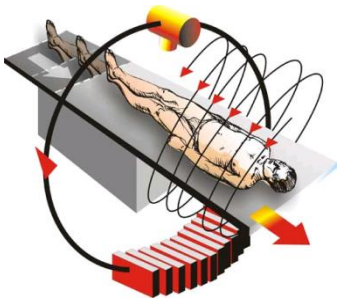


CT

- Clinical

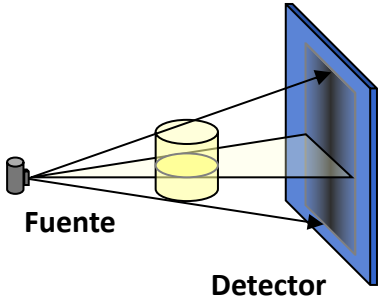


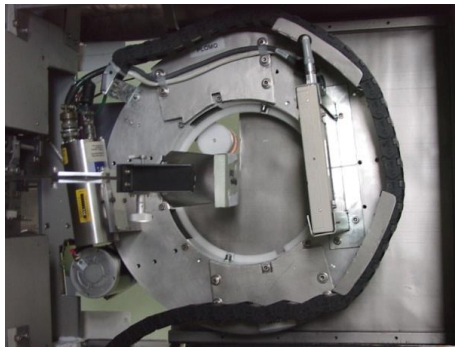






- Small animal





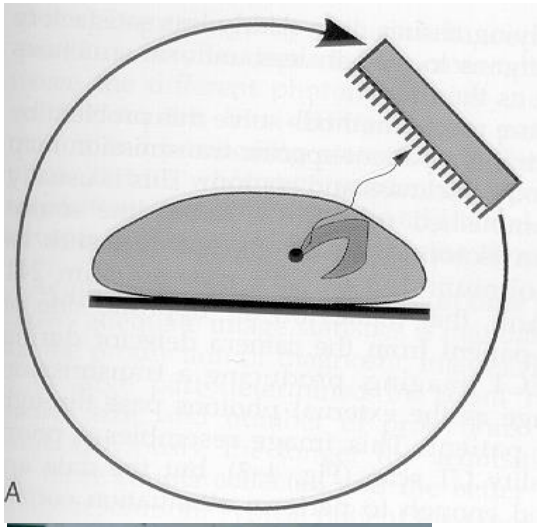






SPECT: Single Photon Emission Tomography

- Clinical



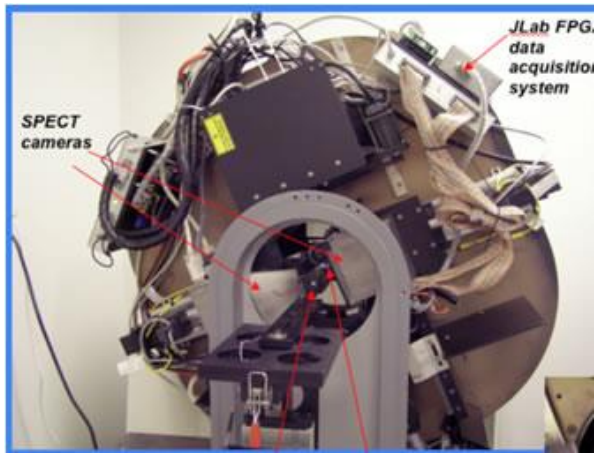


- Small animal

MILabs







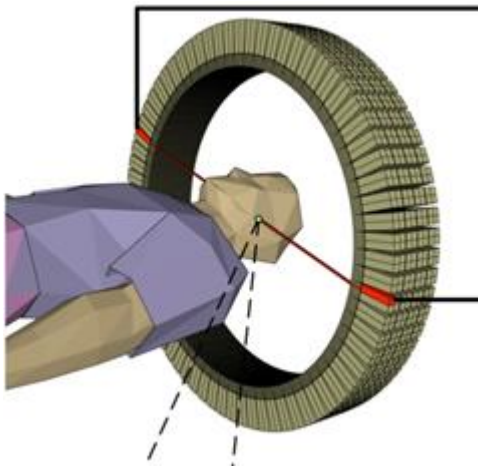
Jefferson Lab

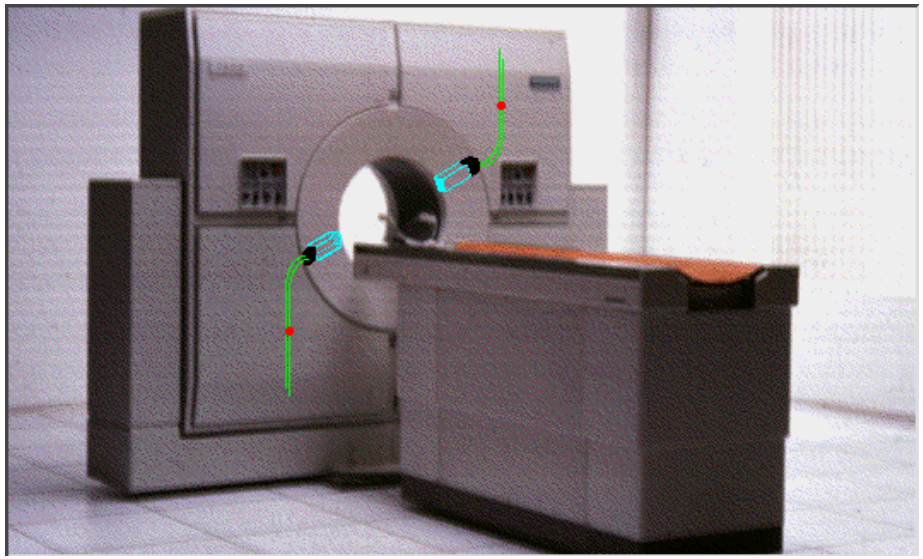
UC3M



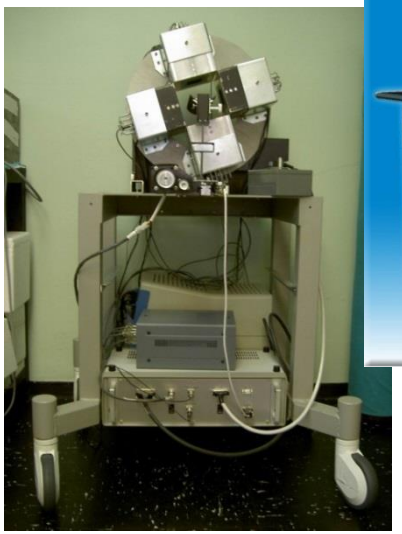
PET: Positron Emission Tomography

- Clinical

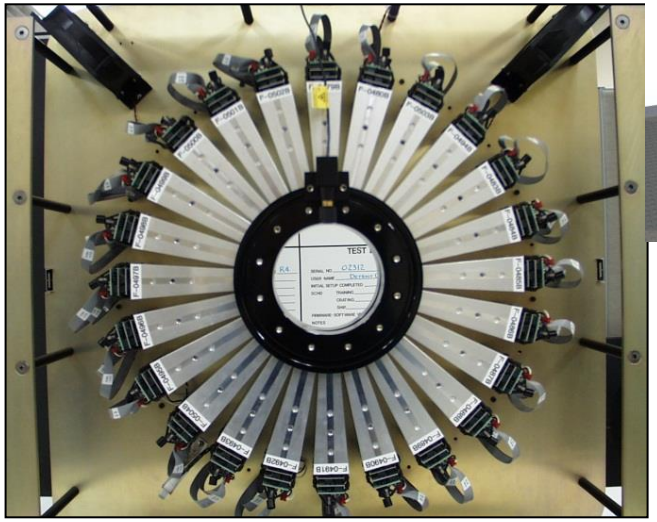




- Small animal





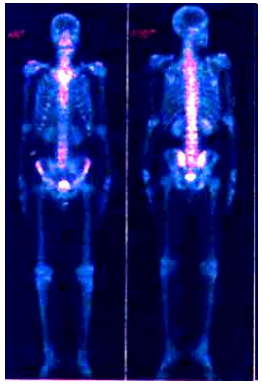


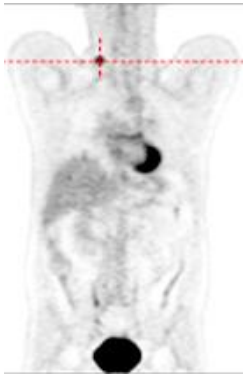


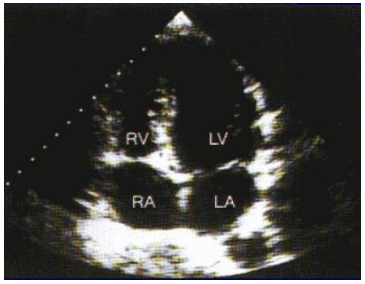
Medical imaging modalities I

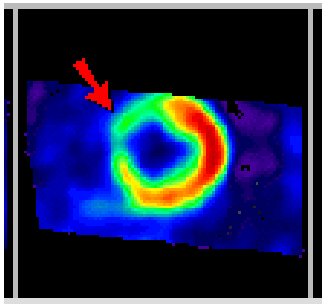
Mónica Abella (monica.abella@uc3m.es)

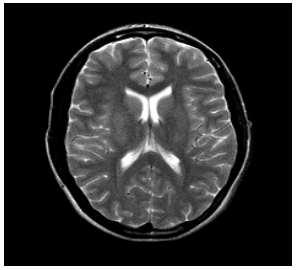




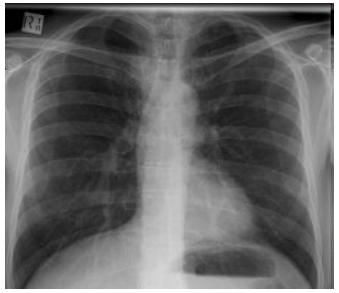


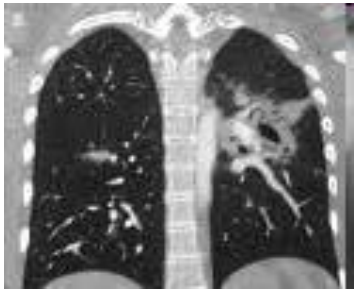


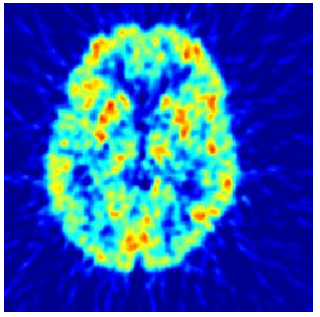








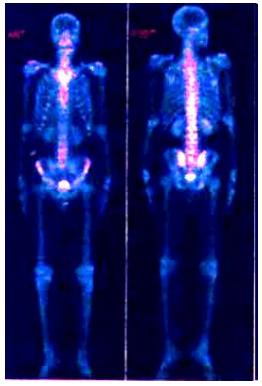


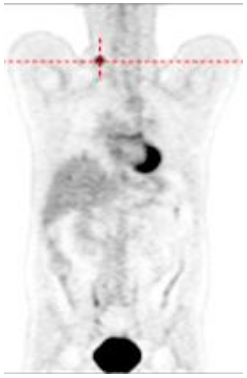


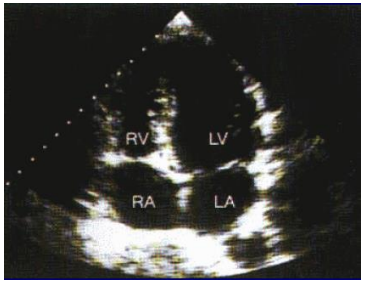
Medical imaging modalities II

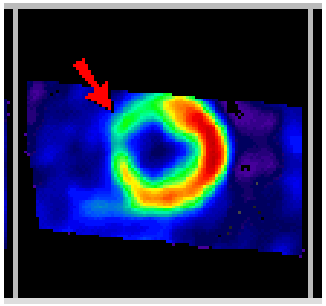
Mónica Abella (monica.abella@uc3m.es)

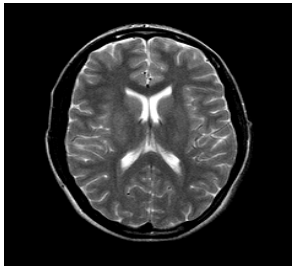




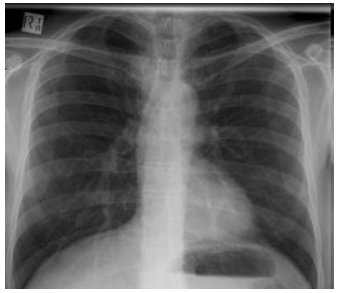


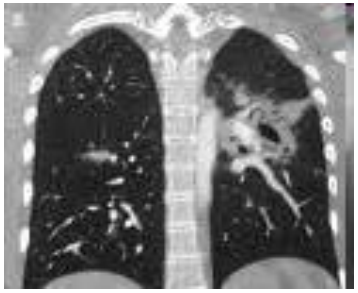


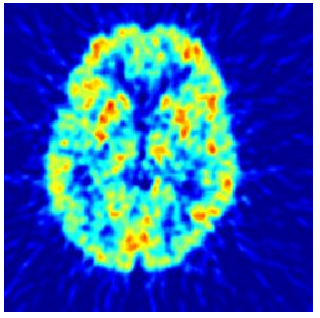








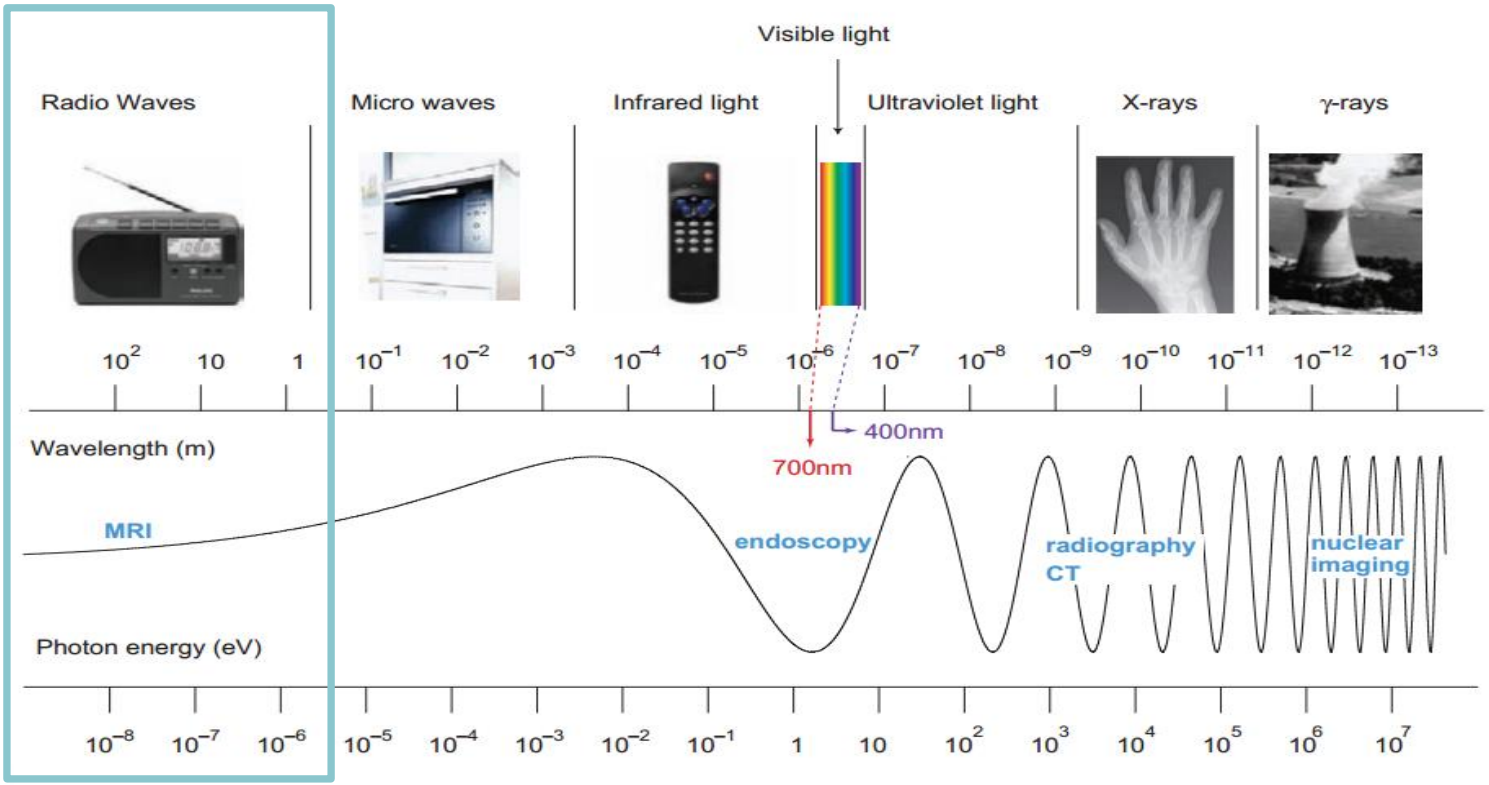




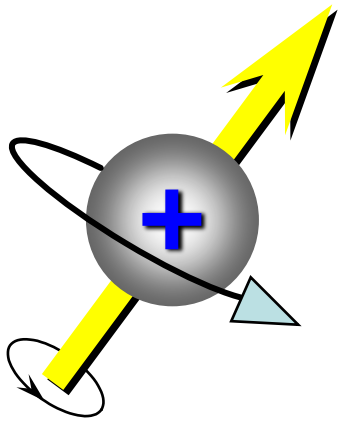
MRI

Energy in MRI

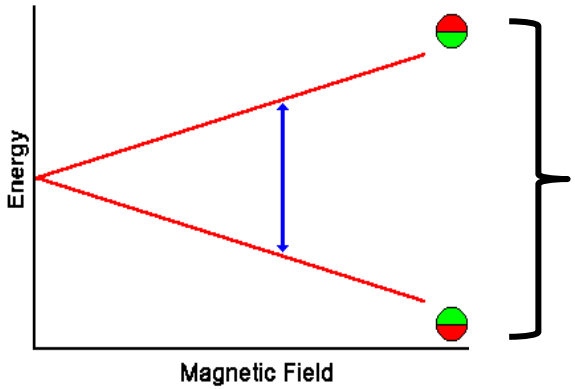
- Type of energy: electromagnetic radiation
  - MRI: radiowaves
- Low energy: non ionizing



Spin within a magnetic field



- Spin: intrinsic form of angular momentum carried by elementary particles
- Many subatomic particles have spin
  - Electrons, protons, neutrons
  - Almost every element has an isotope with overall nuclear spin ≠ 0
- **Inside a magnetic field B**
  - New energy levels (new possible transitions)
  - **Possibility of emission/absorption of low energy photons**

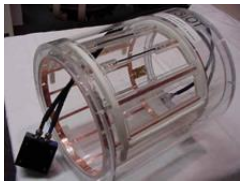


$$\Delta E = h \gamma B$$

Giromagnetic constant

^{1}H	42.58
^{2}H	6.53
^{13}C	10.71
^{14}N	3.1
^{19}F	40.05
^{23}Na	11.26
^{31}P	17.24

Nuclear Magnetic Resonance



RF

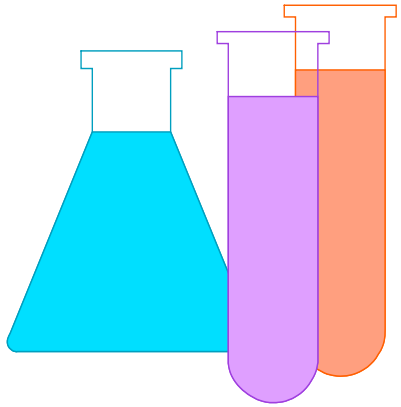
$f = \gamma B_0$

$E = hf = h\gamma B_0$

Giromagnetic constant

^{1}H	42.58
^{2}H	6.53
^{13}C	10.71
^{14}N	3.1
^{19}F	40.05
^{23}Na	11.26
^{31}P	17.24





B_0

$f = \gamma B_0$

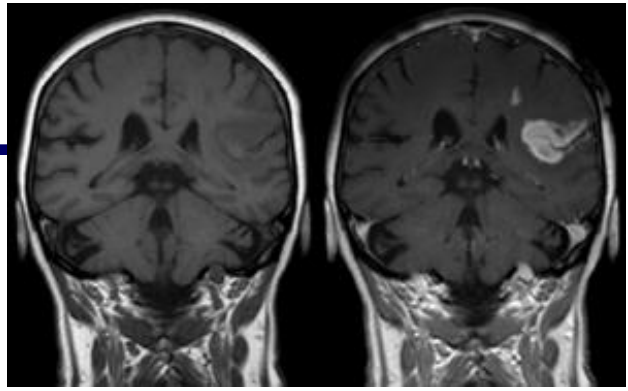
RF

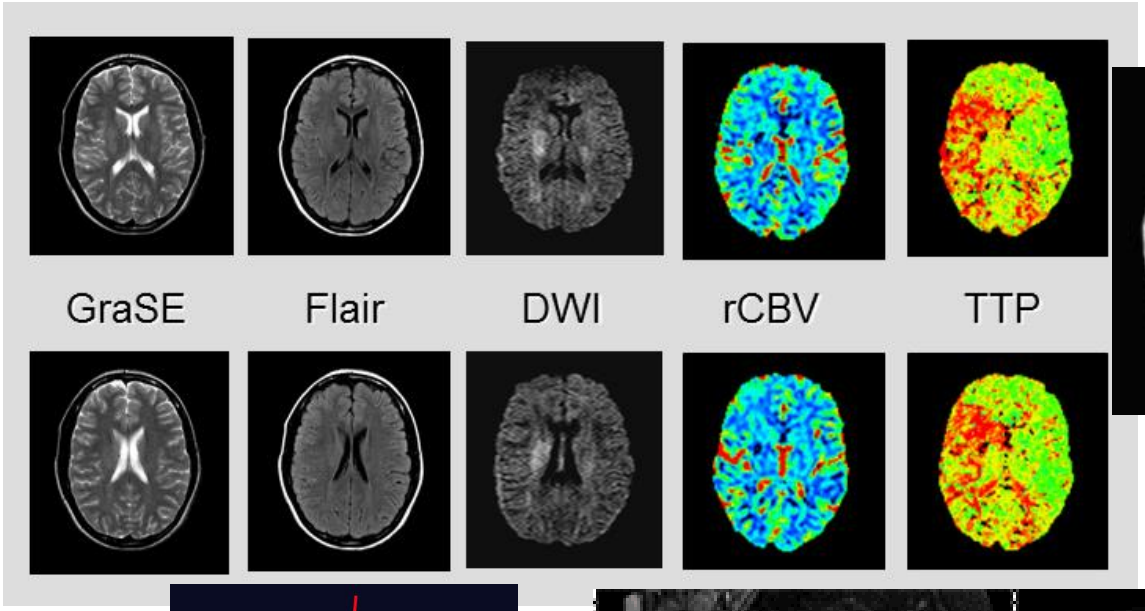
eco

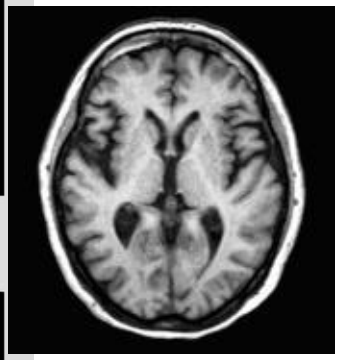


Origin of the contrast in MRI

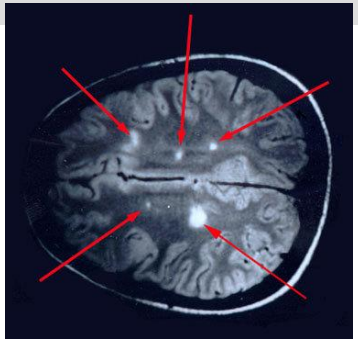
- Hydrogen (water) content results in contrast between tissues
- Many other mechanisms, some based on relaxation

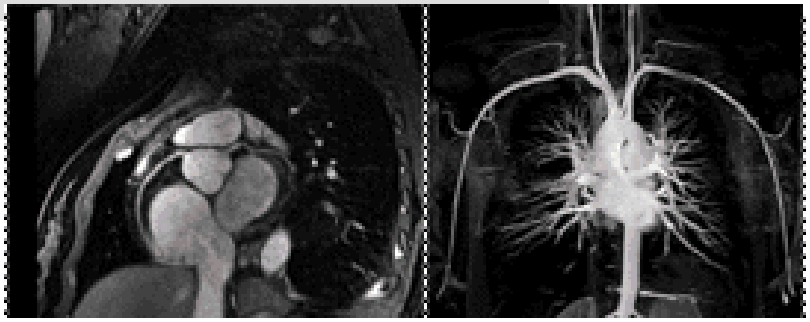








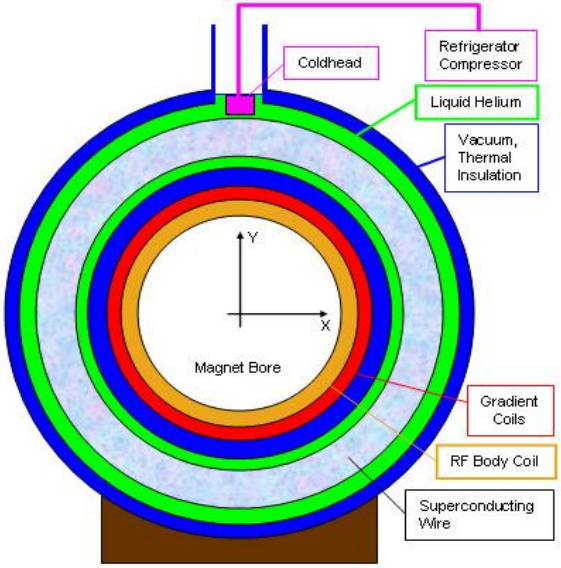


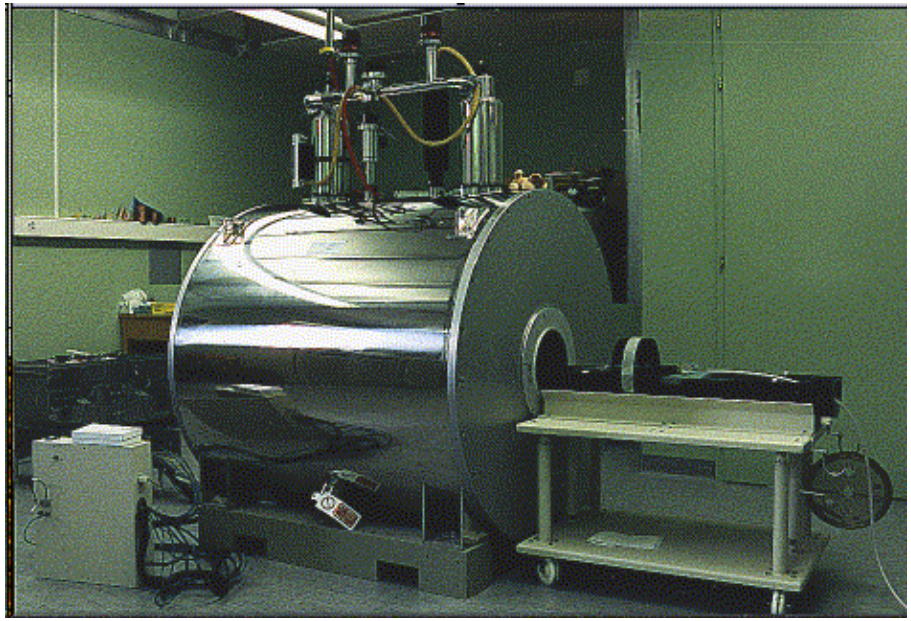




The magnet

- High magnetic field (0.5-7 Tesla)
- Electromagnet made of superconducting cable 0 at 0 kelvin (-273.15° C)
  - Resistance ~ 0 at 0 Kelvin (-273.15° C)
  - Immersed in liquid helium
- https://www.youtube.com/watch?v=6BBx8BwLhqg
- https://www.youtube.com/watch?v=9SOUJP5dFEg







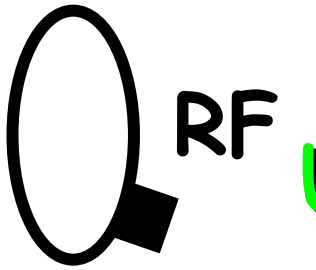
RF antennas and gradient coils







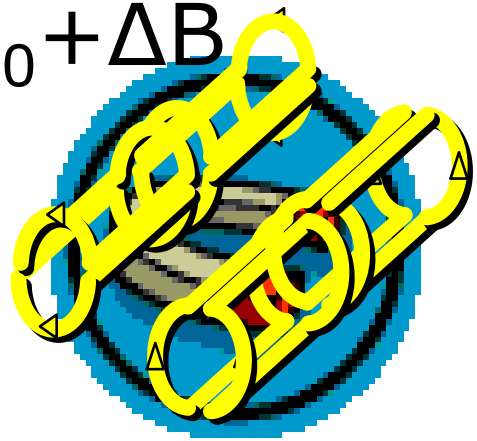
Doing imaging…



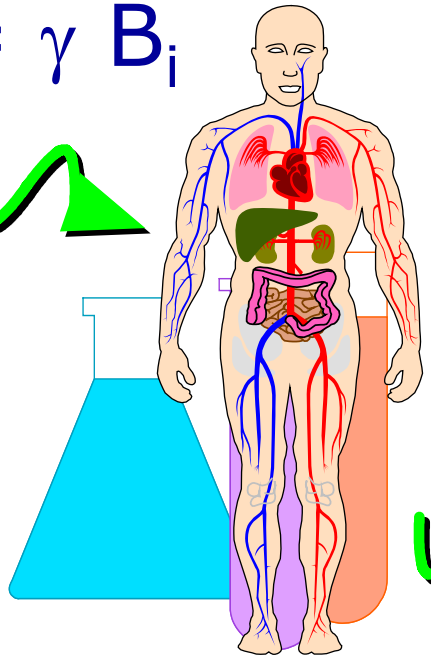
$f = \gamma B_i$

$E = hf = h\gamma B_0$

$B_i = B_0 + \Delta B$

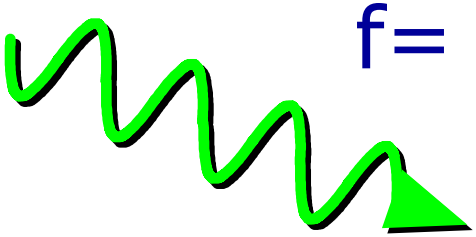


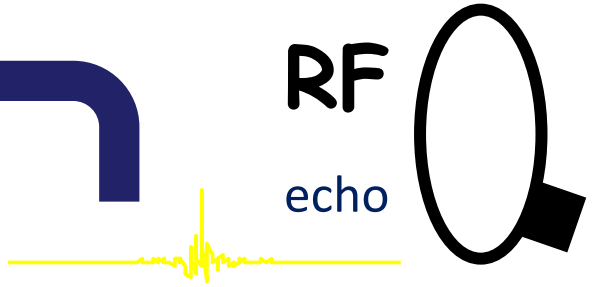
Field gradients

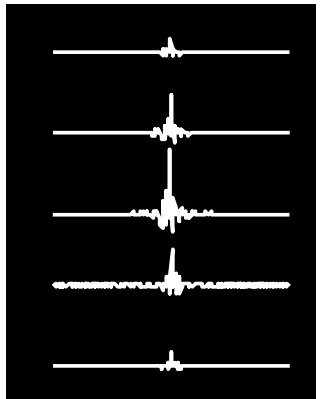


K space

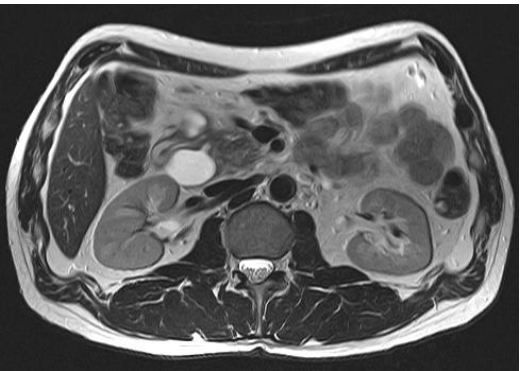
$f = \gamma B_i$



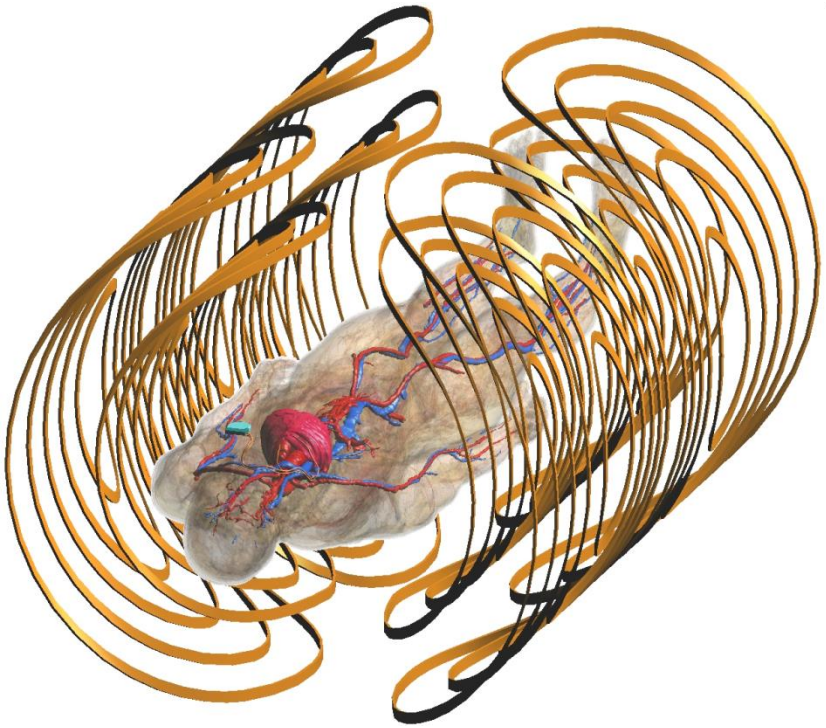


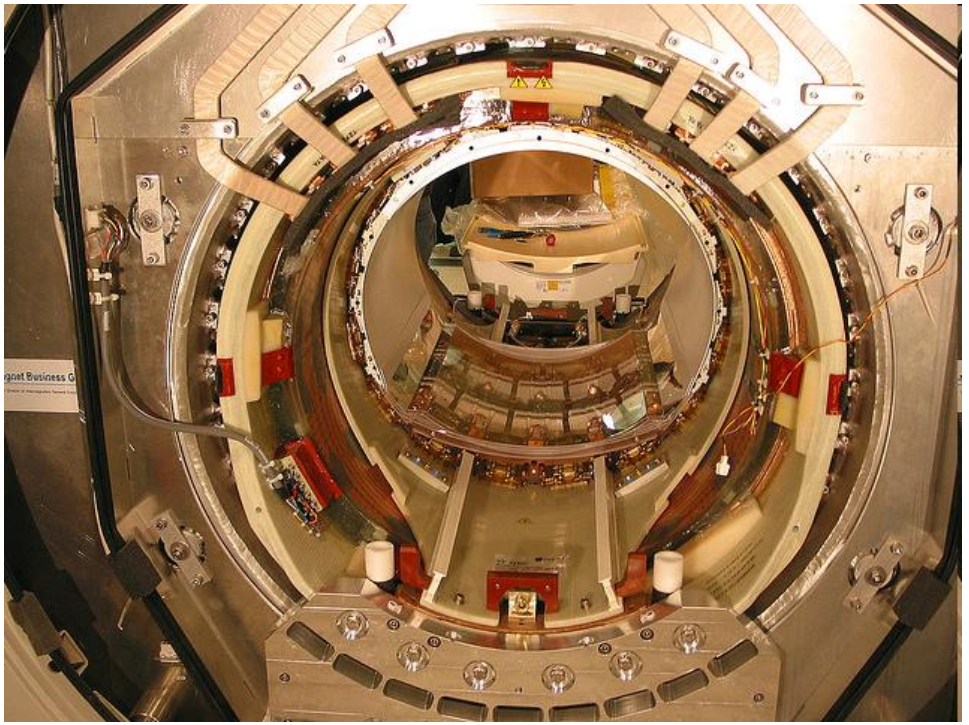


Reconstruction
FT



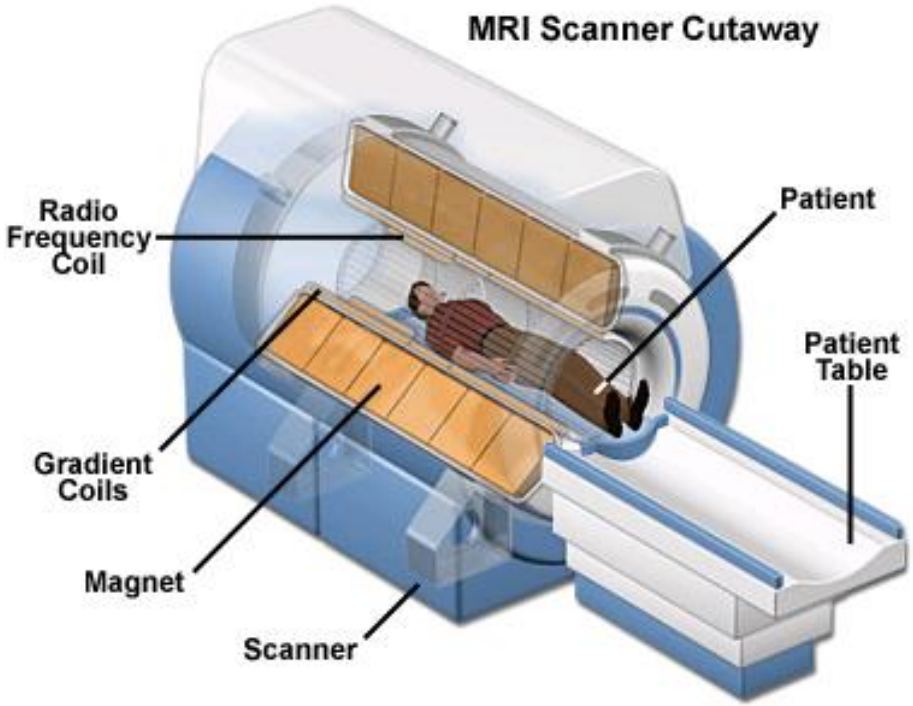
Gradient coils

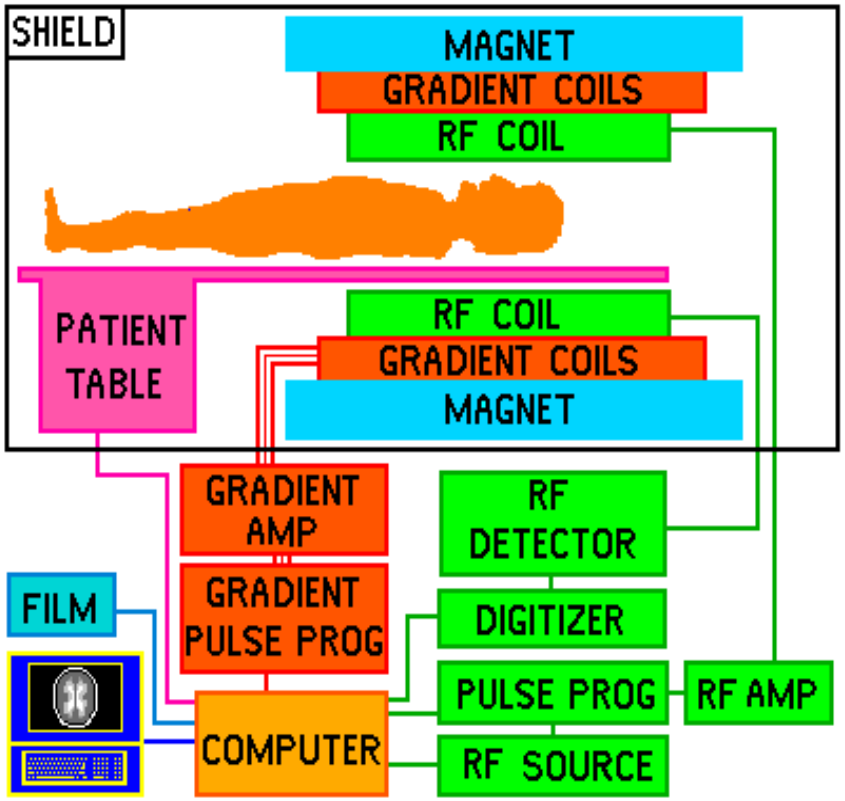




Components of MRI scanners

- The magnet
- The RF subsystem
- The gradient coils
- Electronics
- The reconstruction



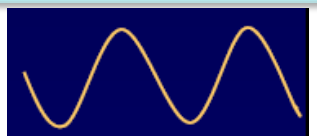


Ultrasound

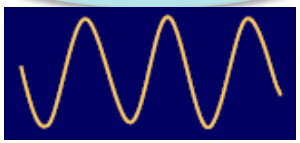
Energy in ultrasound imaging

- Type of energy used: Ultrasound waves
  - Mechanical energy
  - Ultrasonic waves
    - Longitudinal mechanical waves
    - Wave motion: elasticity and inertia of the medium

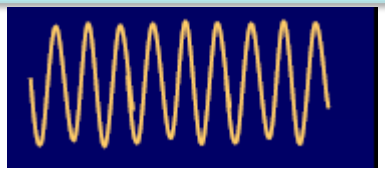
Infrasound < 20 Hz



Audible sound
20 Hz - 20 KHz



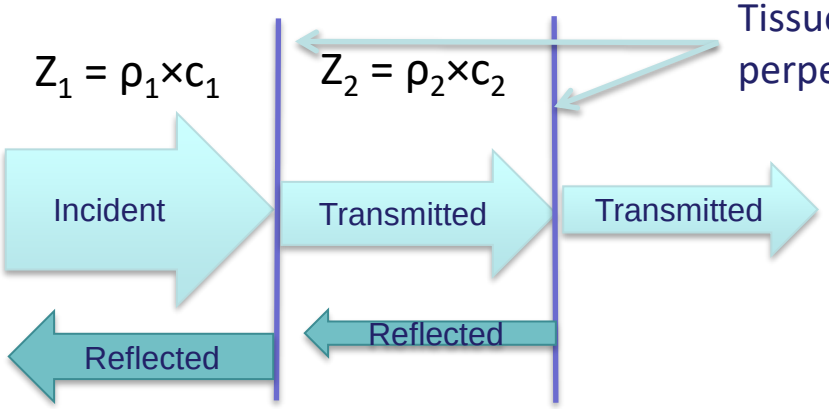
Ultrasound > 20 KHz
Surgical: 25-55 KHz
Diagnostic: 1-10 MHz



- Non-ionizing radiation, inexpensive, portable, excellent temporal resolution

Interaction with tissue in US

- A propagating wave is partially reflected at the interface between different tissues
  - Position of the interface can be extracted from the time delay
  - The intensity of the echo is used to determine the brightness of the image at the reflecting tissue surface



Tissue interface smooth compared to λ and perpendicular to the direction of wave propagation

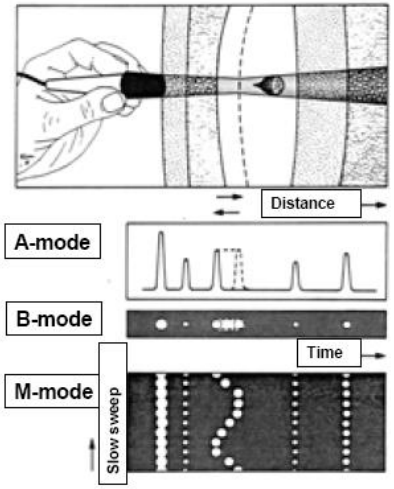
Reflection coefficient

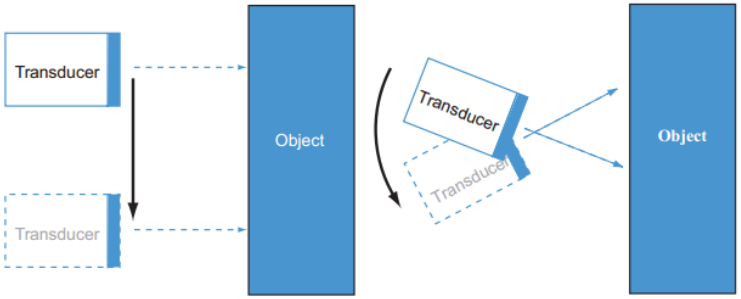
$$I_{reflected} = I_{incident} \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

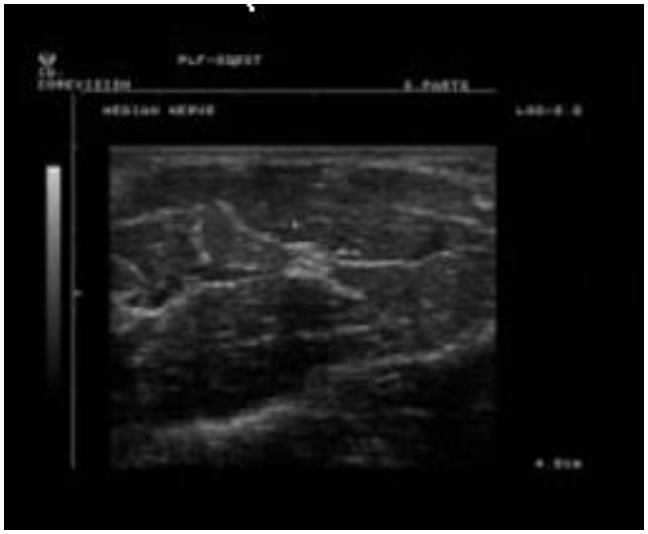
- At soft tissue/bone interface: reflected signal ~40% of the incident intensity
  - imaging of structures deeper-lying than bone is extremely difficult
- At a soft tissue/gas interface: ~99% of the beam intensity is reflected
  - impossible to scan distal structures deeper than lungs or gas-containing bowel

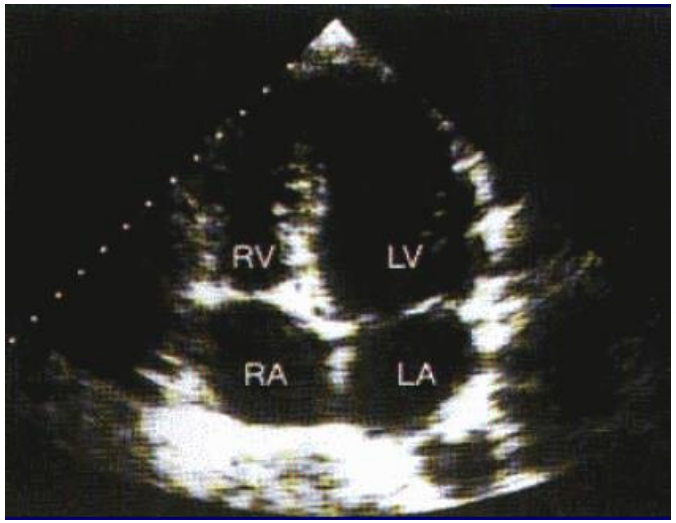
Mode B (Brightness)

- An image is obtained by sending pulses in different directions
- Brightness represents the amplitude of the eco from each point



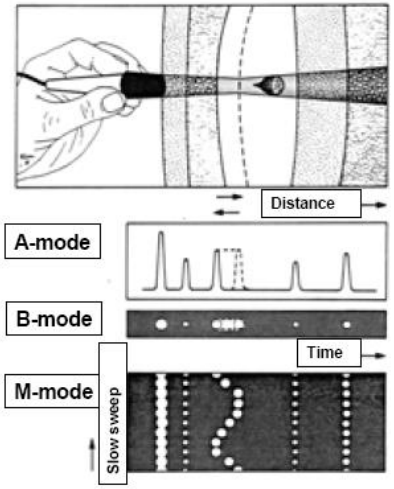


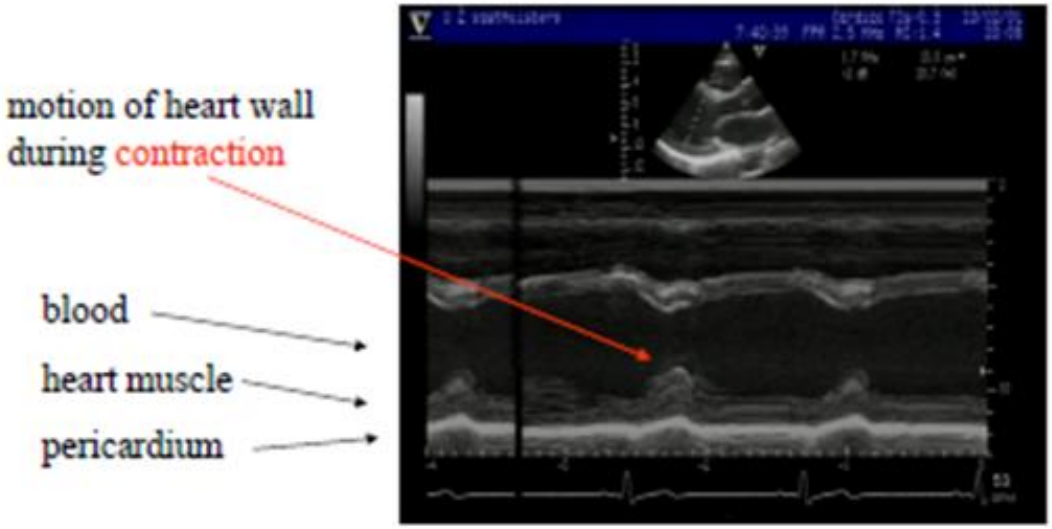


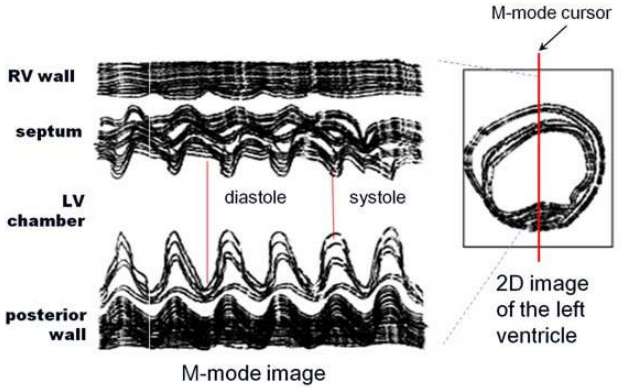


Mode M (Motion)

- Repeated A-mode measurement in time
- Brightness represents the amplitude of the eco from each point
- High sampling frequency: up to 1000 pulses per second
  - Useful in assessing motion
  - Still used extensively in cardiac and fetal cardiac imaging



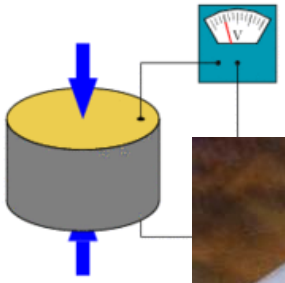




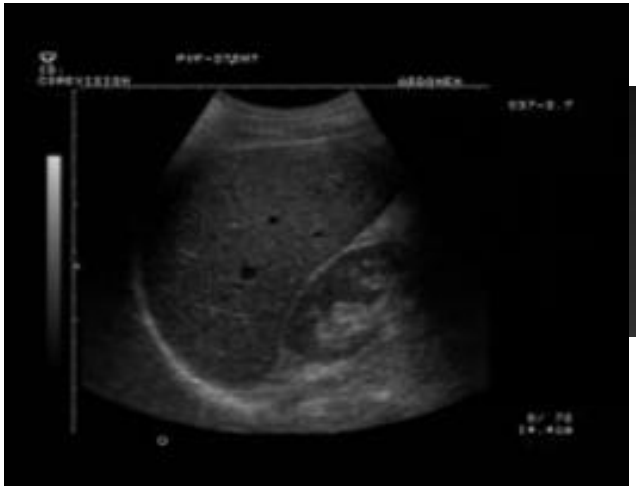
Source & detector in US



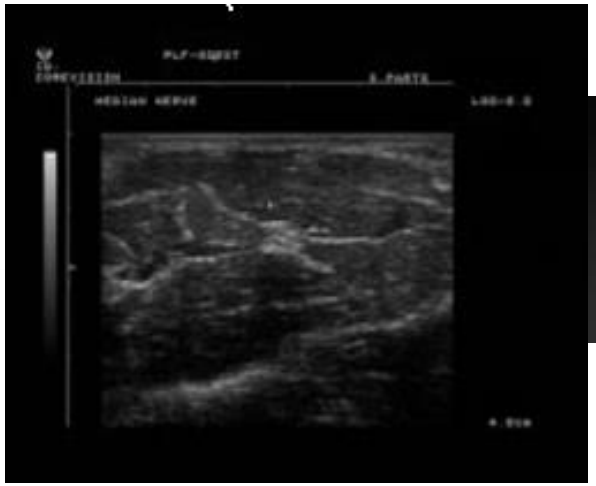
- Source-detector: transducer (converts one type of energy into another)
  - Piezoelectric crystal sandwiched between a pair of electrodes (Pierre Curie,1880)
    - Electricity into sound = pulse
    - Sound into electricity = echo

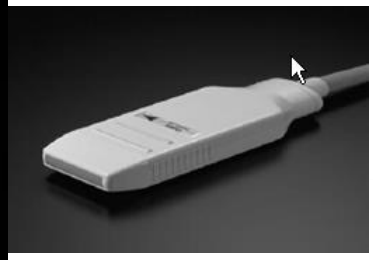












Scanners

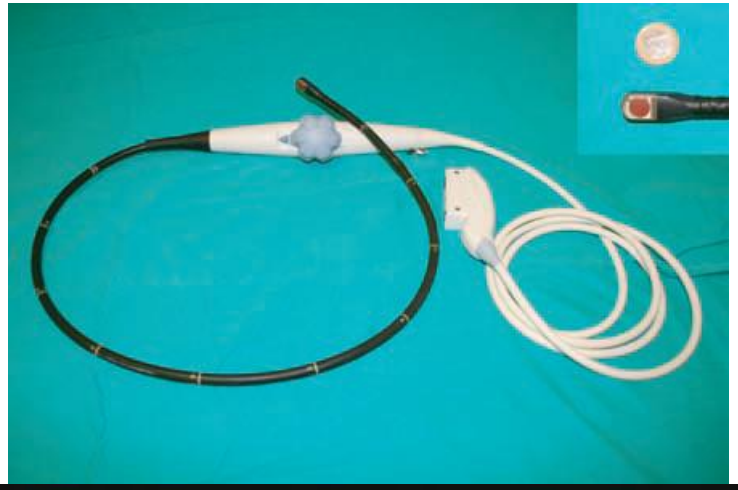




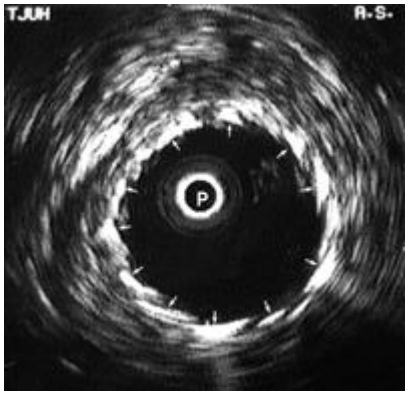




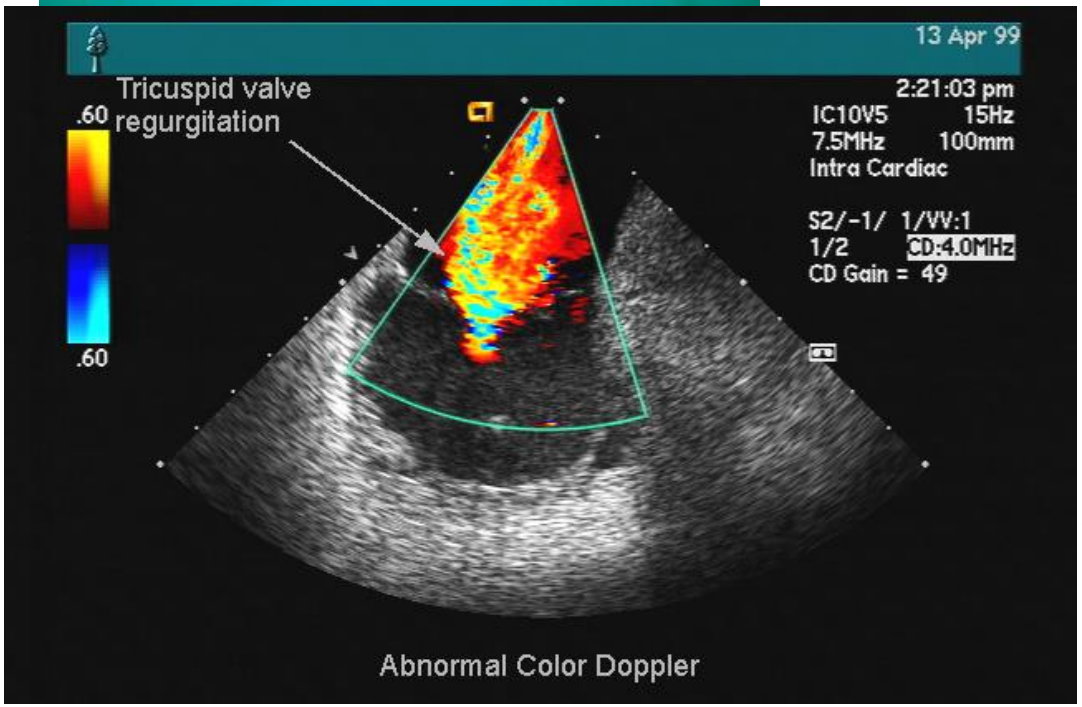
Intraluminal ultrasound

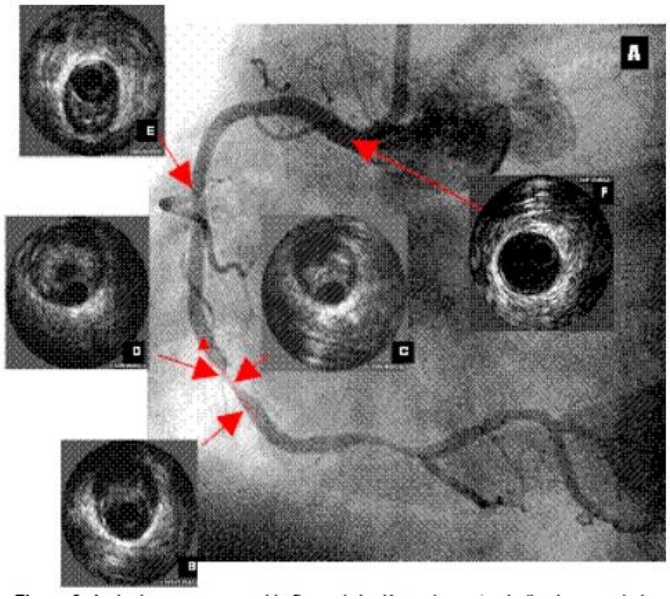






IVUS

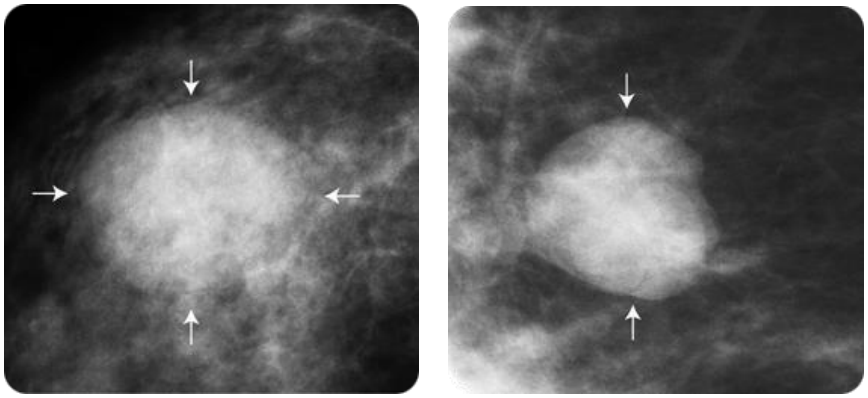




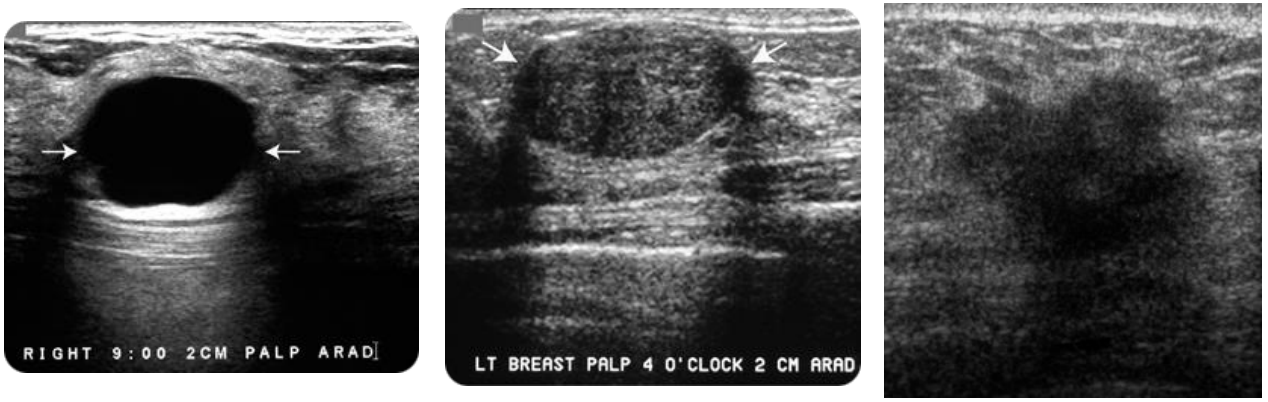
Question

- If I can see anatomy with US and CT, what would you consider to decide which one to use for a patient?
  - Differentiate between cyst, fibrosis, or carcinoma

CT



US



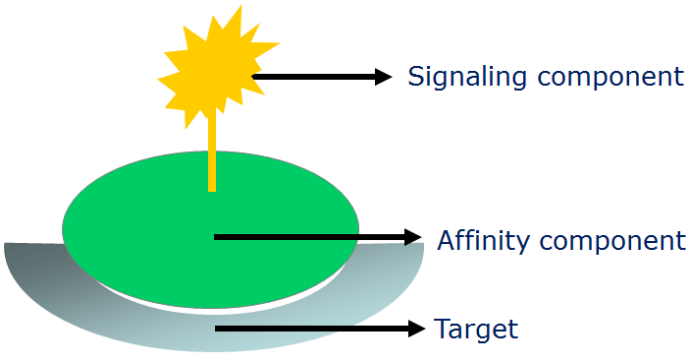
Remember: Contrast agents in X-rays

- X-ray imaging has poor contrast in soft tissue
  - Contrast agents: **High attenuation to X-rays**
- Common contrast agents:
  - Iodine (blood vessels)
  - Barium (digestive tube)
- Administration
  - Intravenous injection (blood vessels)
  - Oral administration (abdomen and pelvis)
  - Rectal administration (large intestines and other organs in the pelvis)



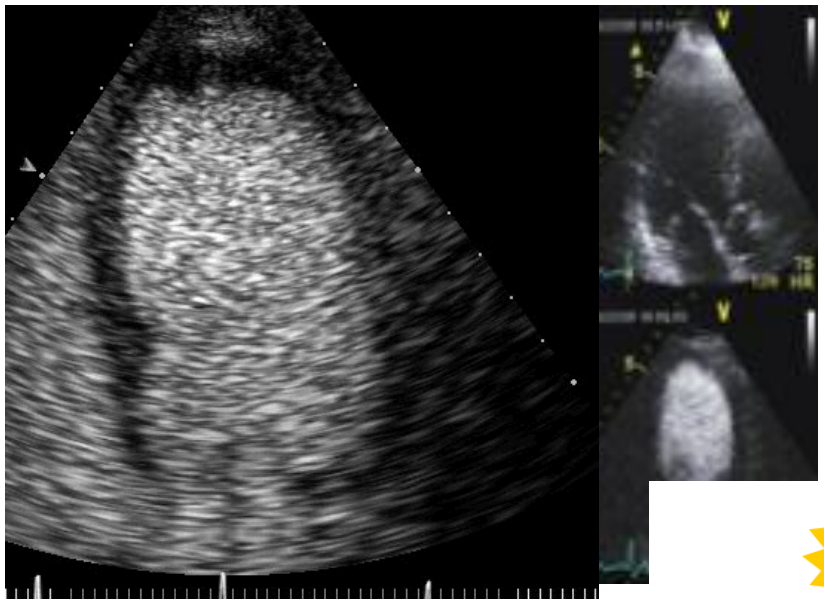


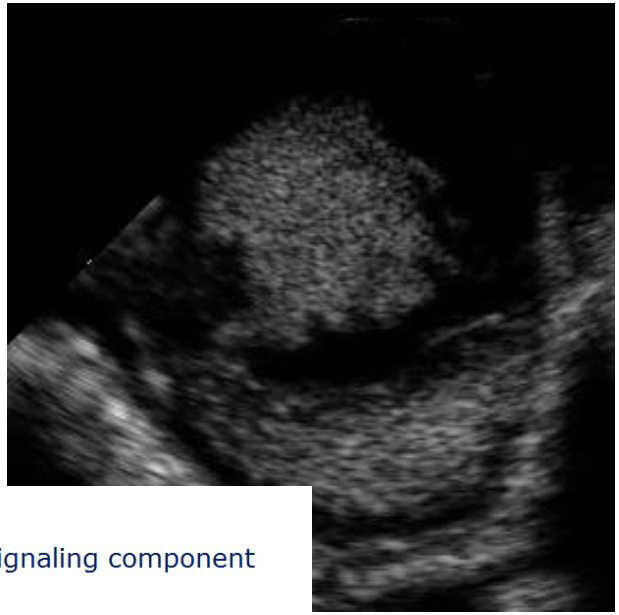

Is it a tracer?



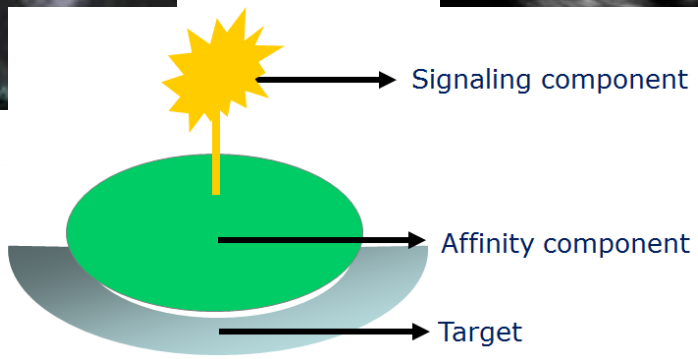
Contrast agents in US: Microbubbles

- Air-filled microbubbles administered intravenously
- High degree of echogenicity: ability of an object to reflect the ultrasound waves
- Blood perfusion in organs, blood flow rate in heart, …





Is it a tracer?

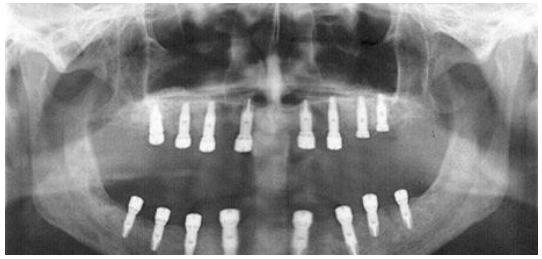


Clinical applications

Planar X-Ray Applications

- Bone radiography:
  - Fractures, joint luxation, spine scoliosis
  - Lesions, infections, arthritis, abnormal bone growth (exotosis)
  - Bone cancer
- Dental











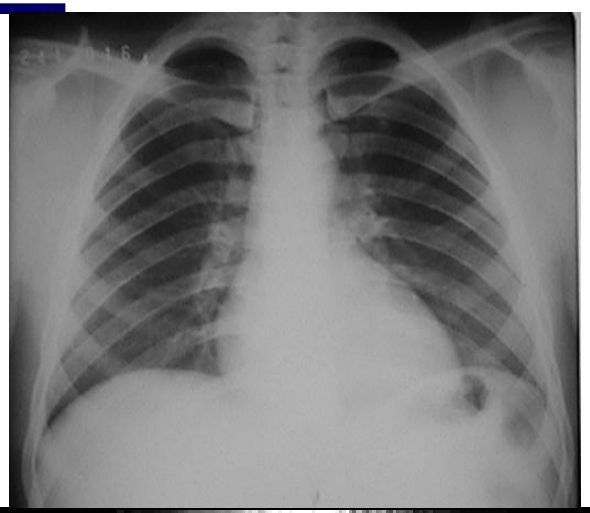


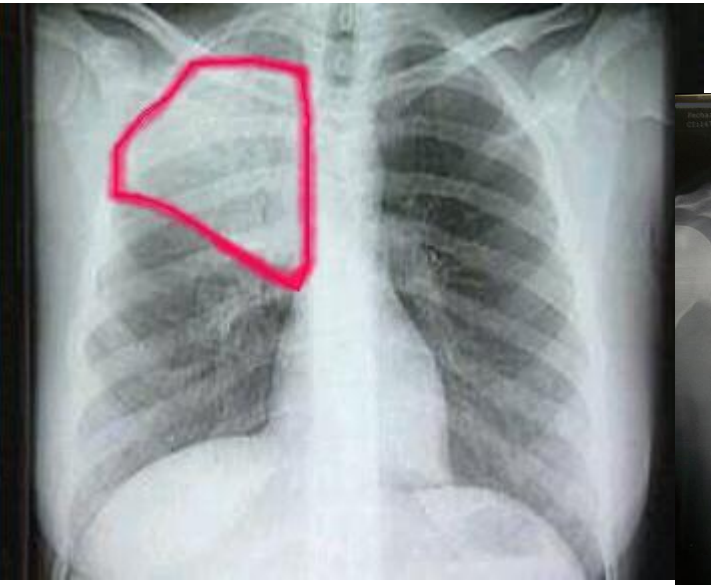




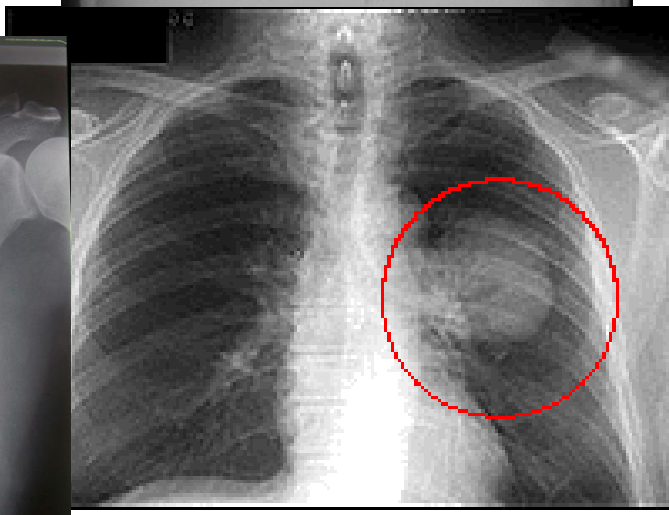
Planar X-Ray Applications

- Thorax
  - Lung: pneumonia, lung edema, lung cancer
  - Heart diseases (cardiomegaly)







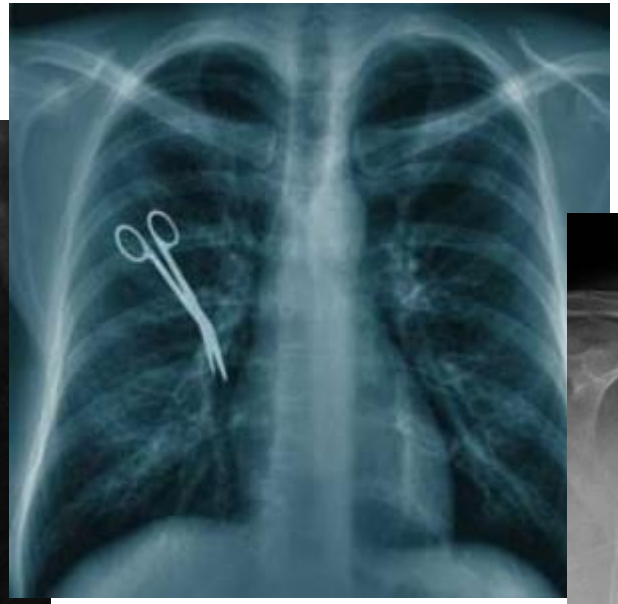


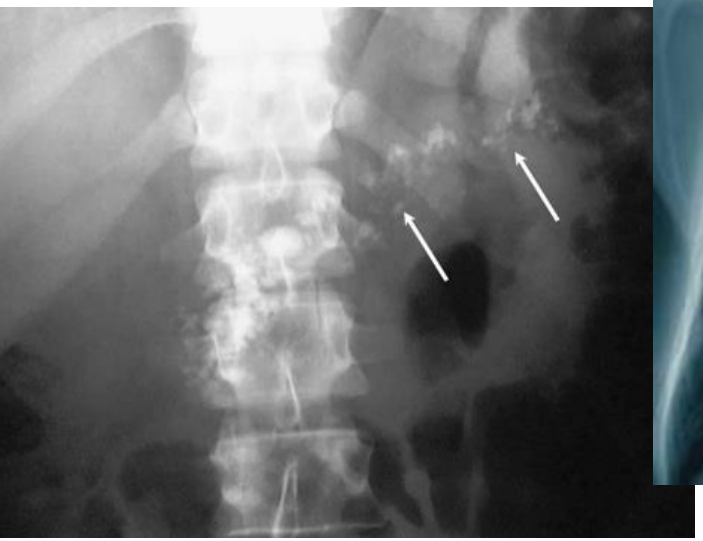
Planar X-Ray Applications

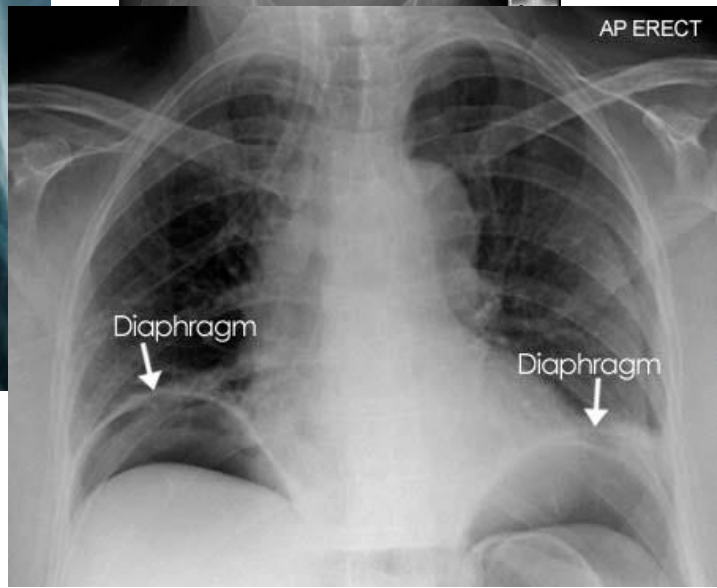
- Abdomen
  - Free air (perforation) and ascites (free fluid in the perit
  - Intestinal obstruction
  - Renal calculus (kidney stone)
  - Gallstone



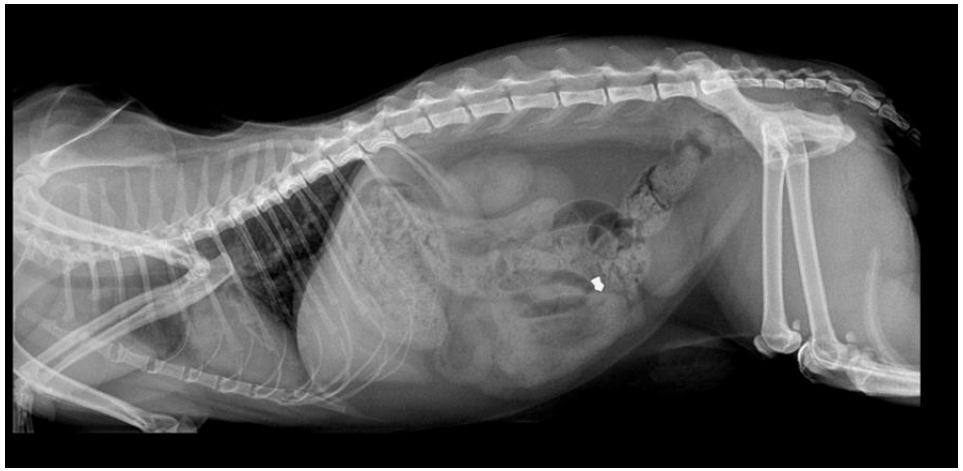


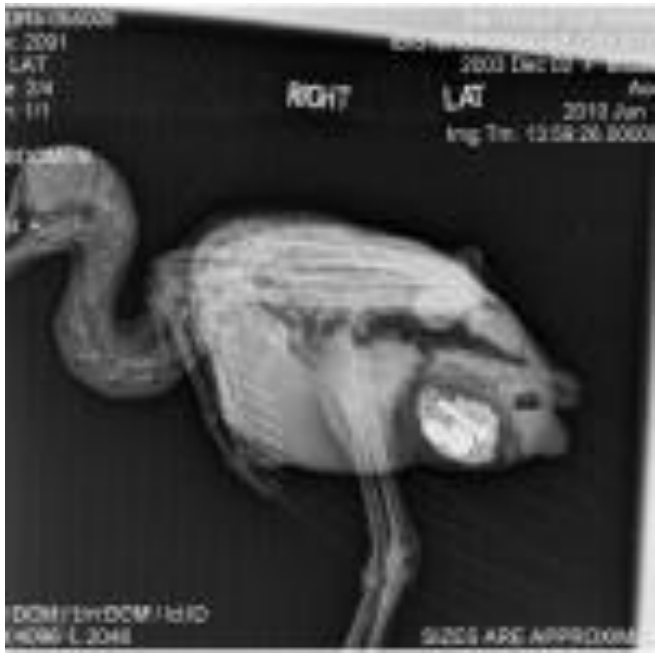






Planar X-Ray Applications

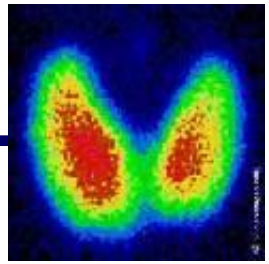


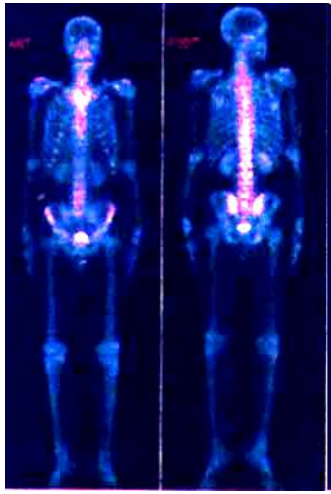


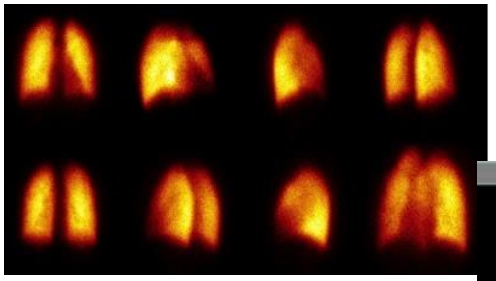


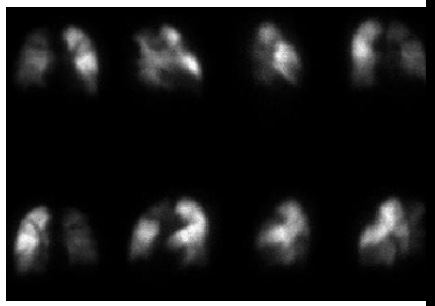
Scintigraphy uses

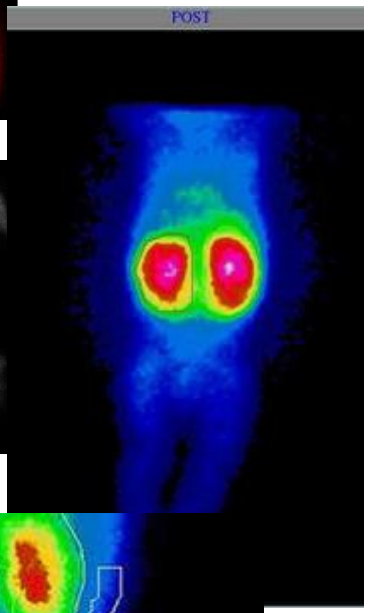
- Bone scintigraphy
  - Small functional alterations before they are visible in a radiography
  - Bone metastasis
- Lung scintigraphy
  - Artery obstruction or thrombus
  - Pulmonary embolism (ventilation/perfusion scan)
- Thyroid scintigraphy
  - Size increase (tryroditis)
  - Nodules
- Radioisotope renography/ Renal scintigraphy



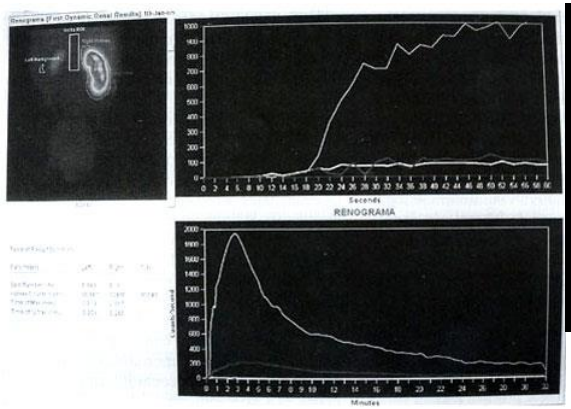


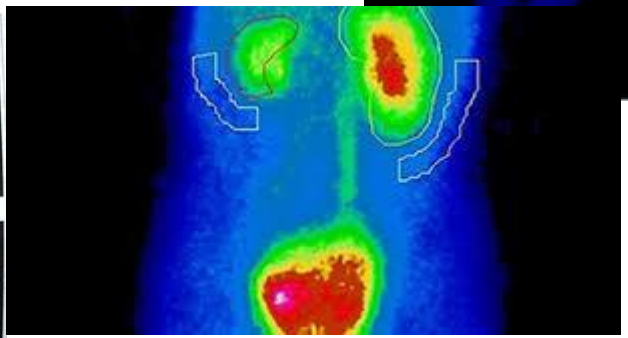








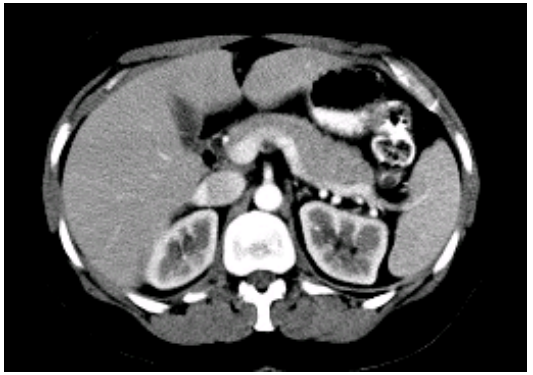


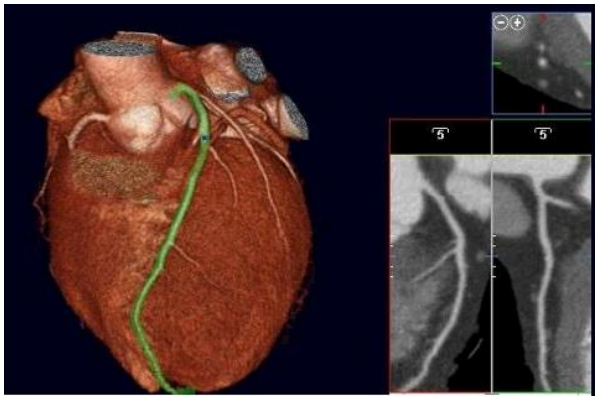


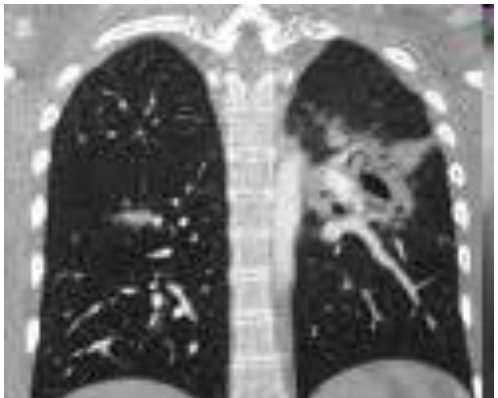
Clinical Uses of CT

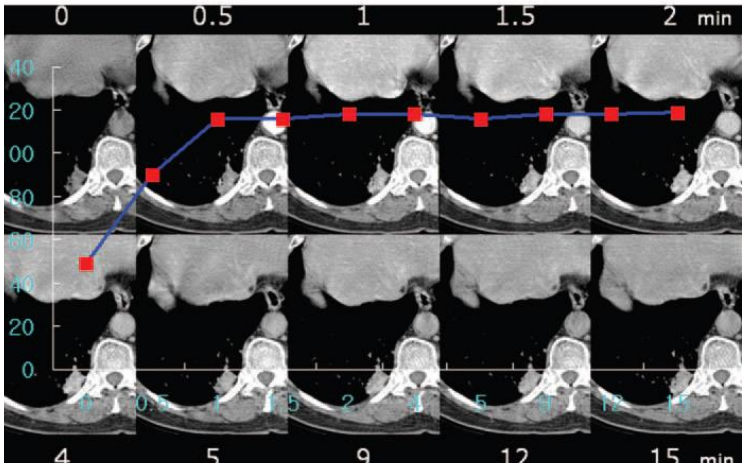
- Not much for bone
- For soft tissue: contrast agents
- Fine detail in lung
  - Difficult diagnosis: dynamic image (vascularization)
  - Emphysema quantification: under research (textures)
  - For screening?
- Skull (trauma, stroke)
- Neck (tumors)
- Abdomen: kidney, liver, etc.
- Heart: dynamic cardiac CT





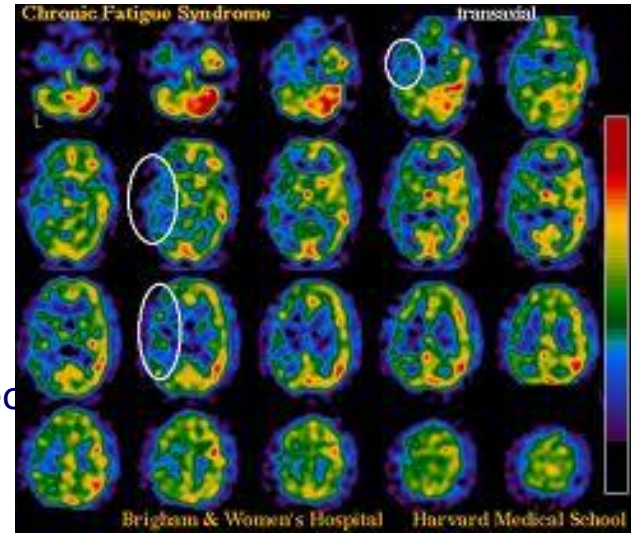


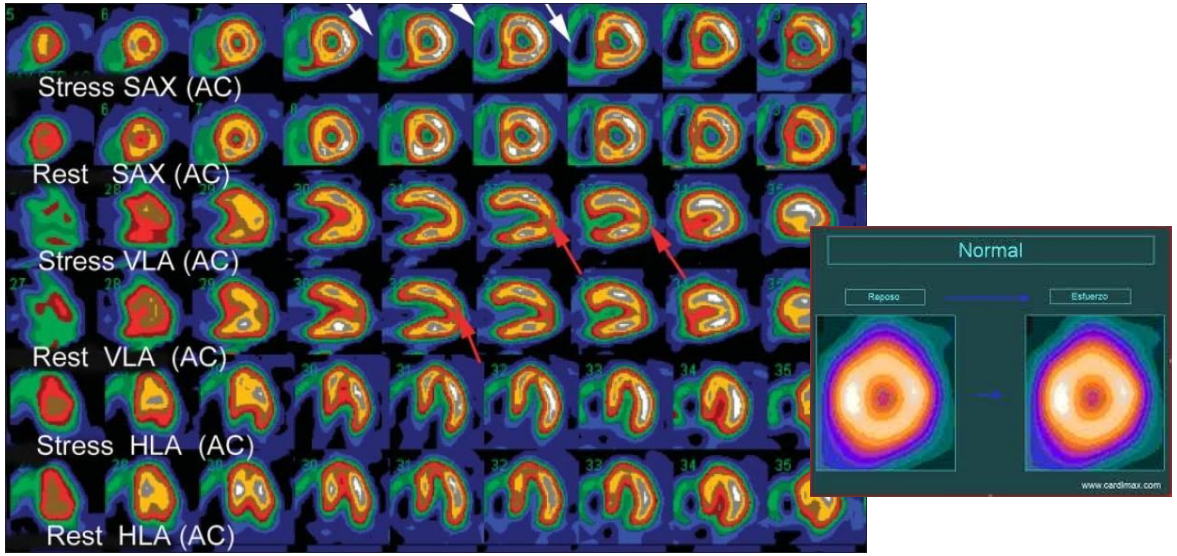


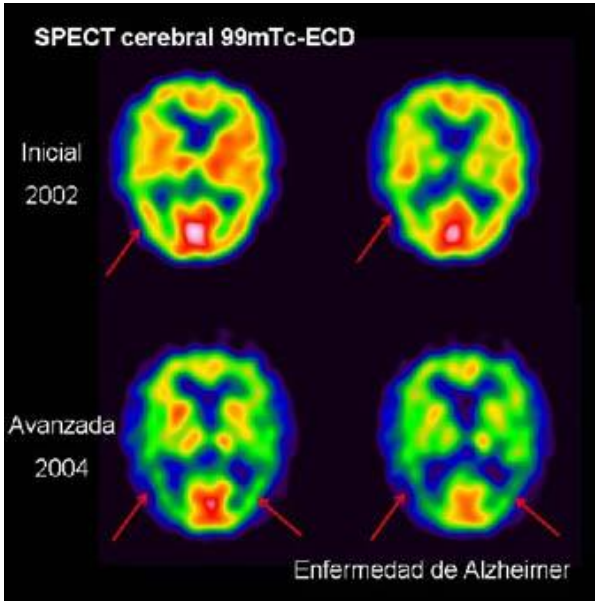


Uses of SPECT

- Brain SPECT
  - Perfusion (blood flow): brain functioning
  - Alzheimer, dementia, epilepsy
- Cardiac SPECT: Myocardium perfusion
  - Rest: death muscle (myocardium infarction)
  - Stress (physical or pharma stimulus): it allows detec
    blood flow (coronary ischemia)
  - Selection of patients for catheterism







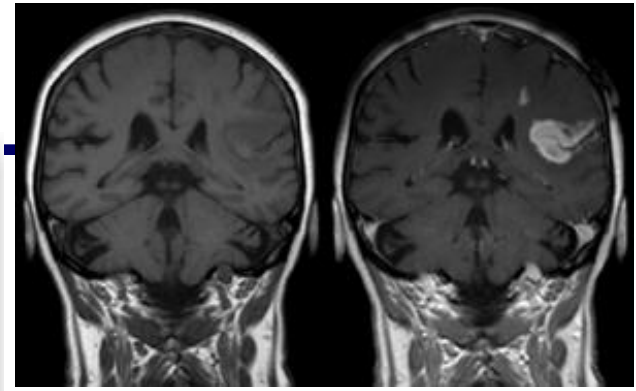
Uses of PET

- Oncology
  - Metastasis and strength of tumors
  - Treatment response (surgery, radiotherapy, Chemother
- CNS
  - Brain function
- Cardiology
  - Coronary illness
  - Evaluation of coronary flow and reserve
  - Tejido viable / no viable
  - FDG y NH_3: Metabolism and flux
- Better image quality than SPECT

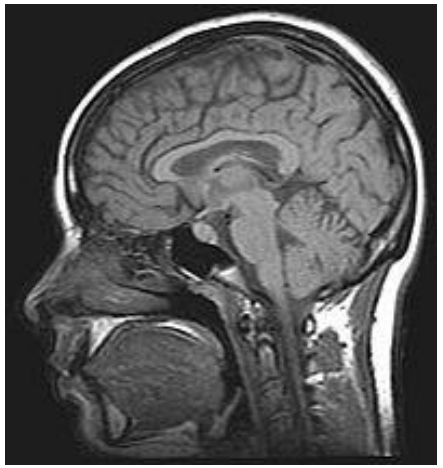


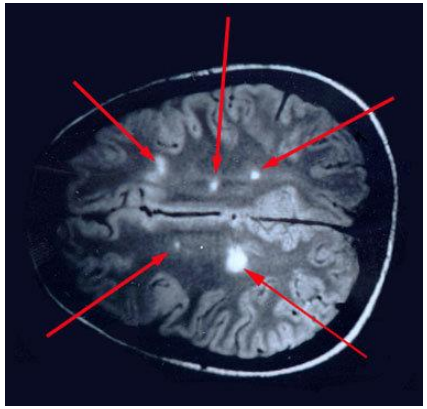
Uses of MRI

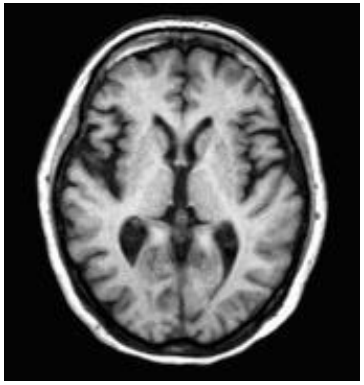
- Contrast agents:
  - Gadolinium (paramagnetic)
- Neuroimaging
  - Clinics
  - Research
- Soft tissue damage (muscle)
- Cardiology
- Angiography



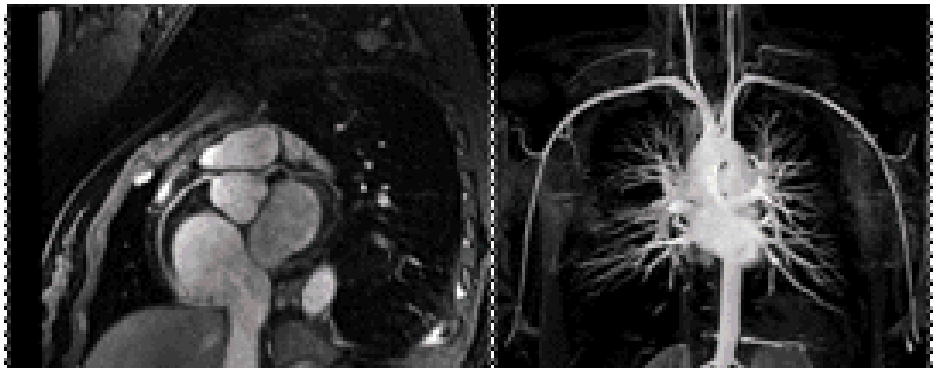








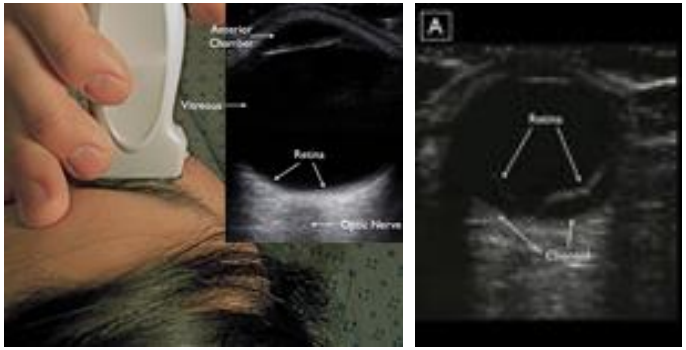


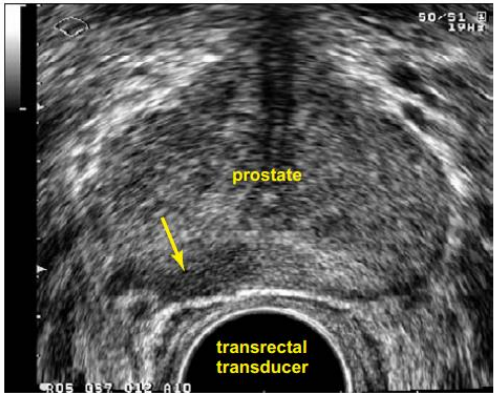


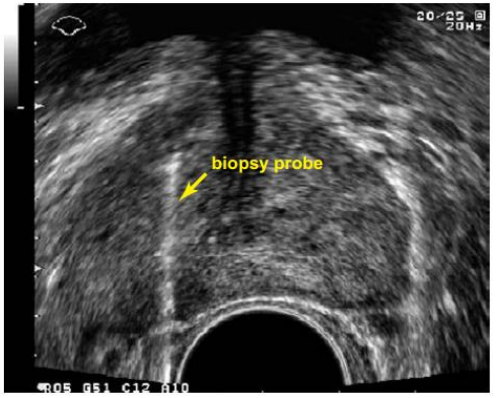


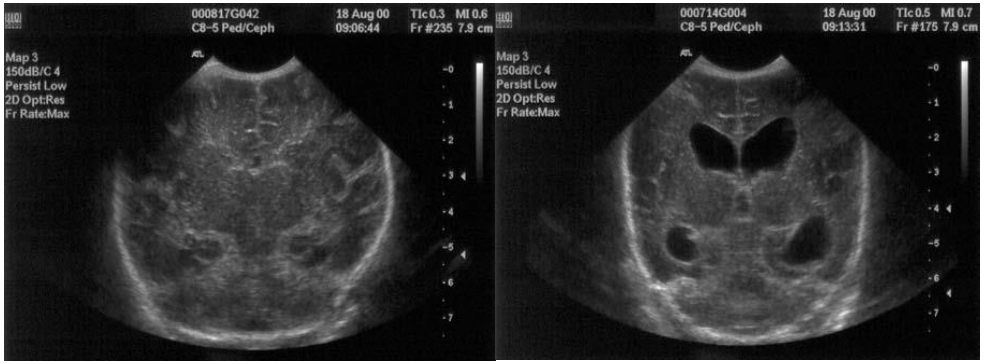
Uses of US

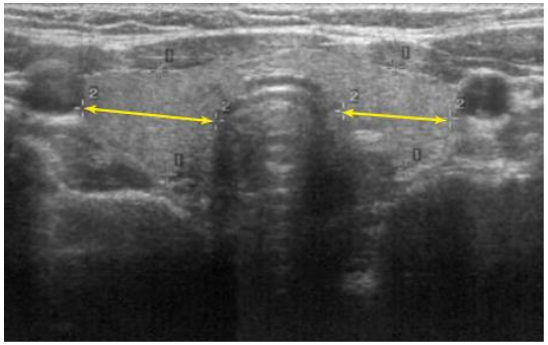
- Head: newborn, retina
- Neck: thyroid, salivary glands, lymph nodes
- Urogenital tract: kidney, bladder, prostate, testicles, uterus, vagina, and ovaries





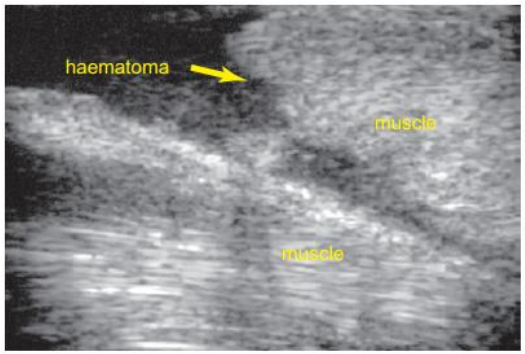


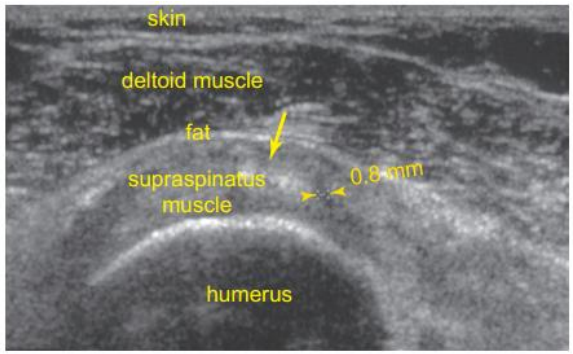


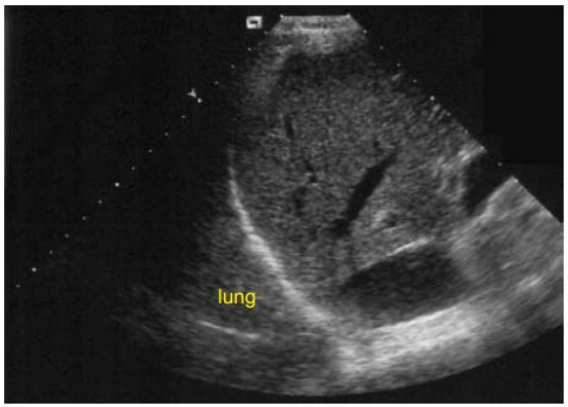


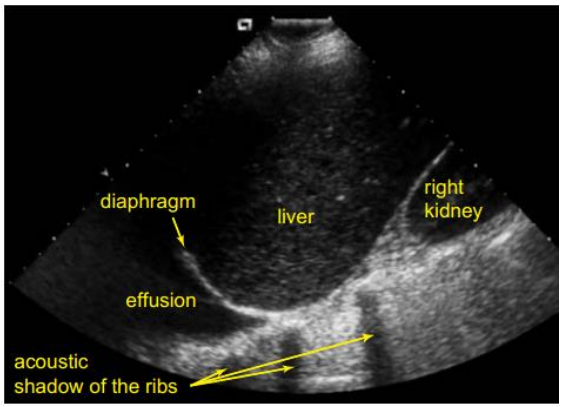
Uses of US

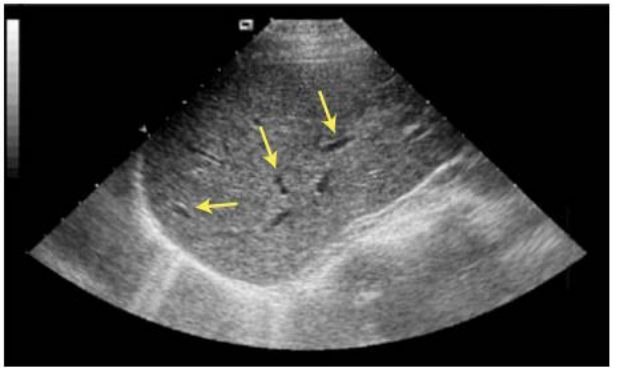
- Musculoskeletal system
- Abdomen: spleen, pancreas, liver
- Thorax: pleura, diaphragm

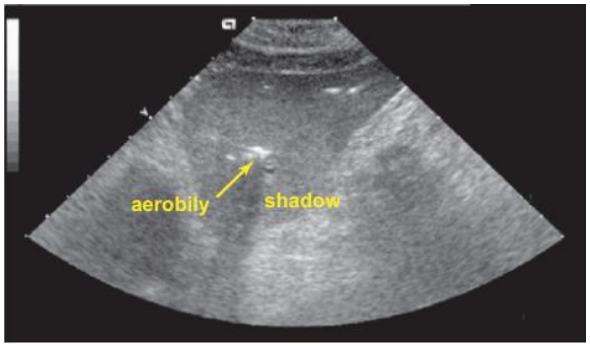


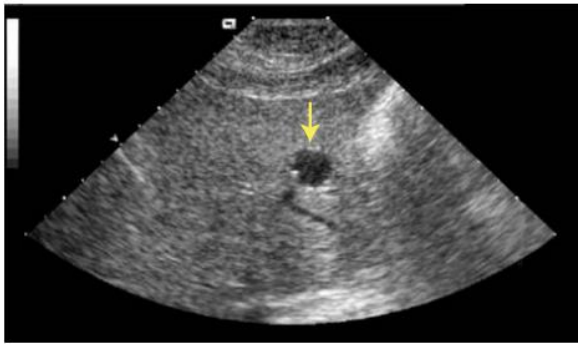










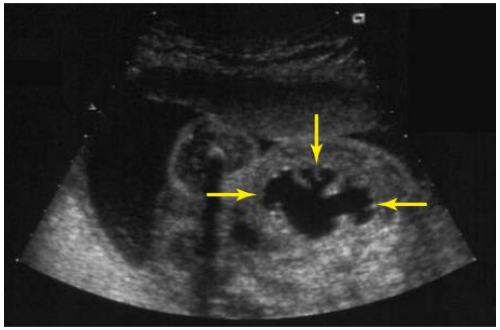


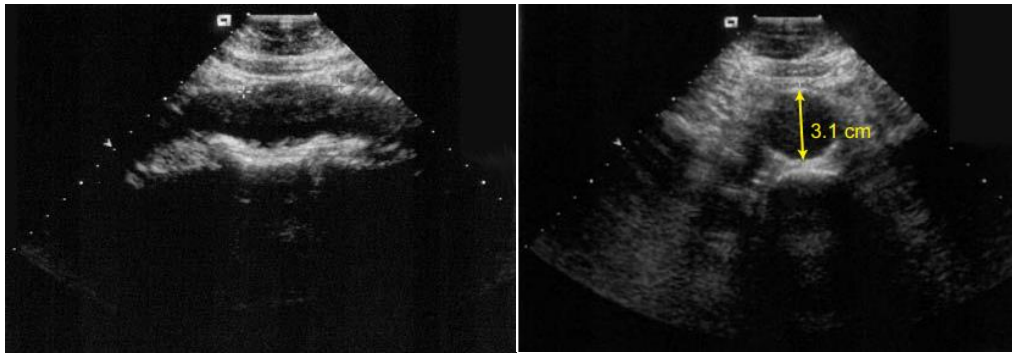
Uses of US

- Cardiology: Echocardiography
- Other fields: anesthesiology (near nerves), vascular studies, breast, nephrology, endocrinology
- Fetus

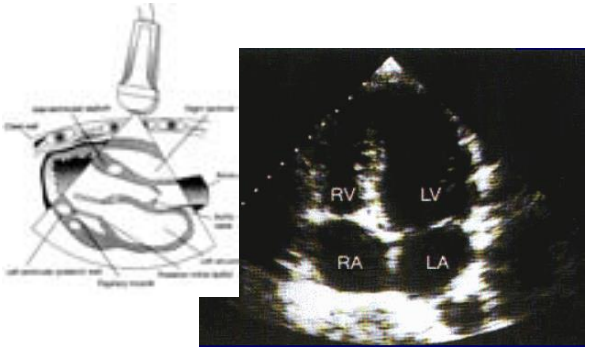


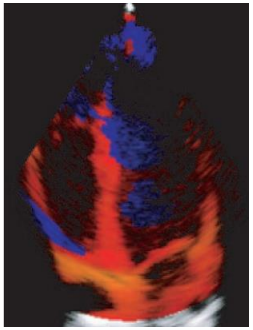


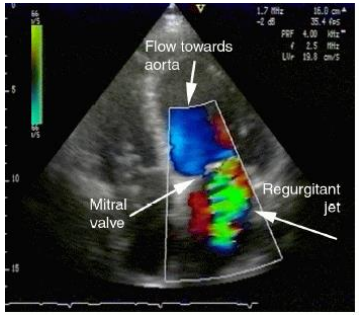






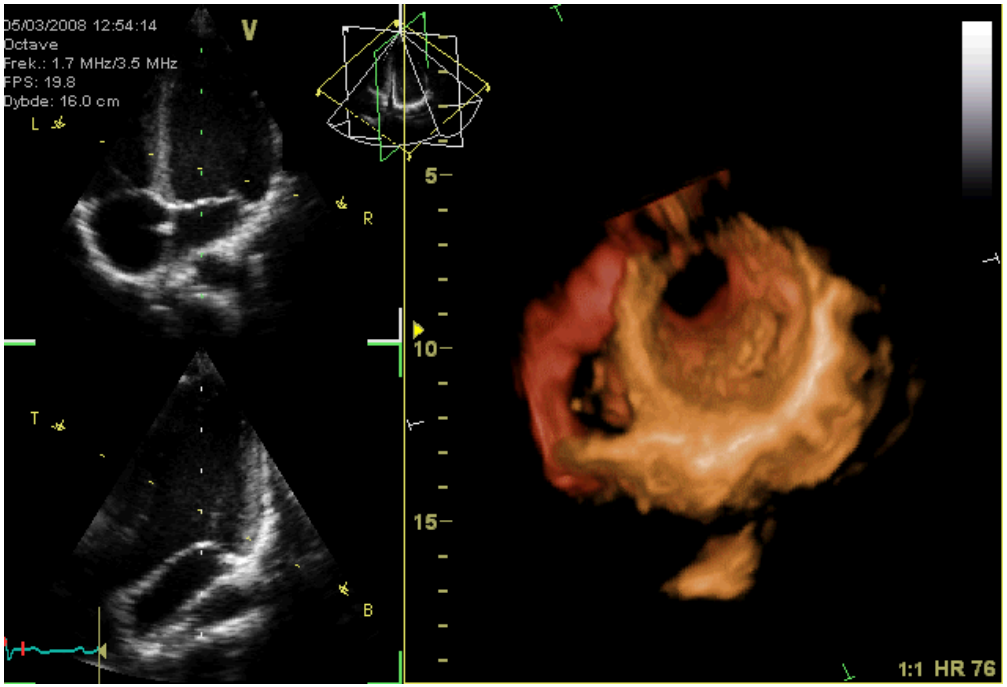






Uses of US

- **Cardiology**



- **Obstetrics**

